
UNIT 4 ESTIMATION OF TWO-VARIABLE REGRESSION MODELS*

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4.0 OBJECTIVES

After reading this unit, you will be able to:

- specify the general form of a bivariate regression model;
- outline the principle behind the OLS method of estimation;
- enumerate the assumptions of OLS method;
- explain the OLS estimation method;
- list the properties of OLS estimators with theoretical details;
- illustrate the interpretation of OLS estimates;
- state the expressions for the standard error of OLS method;
- comment on the characteristics of OLS estimators;

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- prove that the OLS estimators are the best linear unbiased estimators (Gauss Markov theorem);
- outline the significance of the coefficient of determination (r^2); and
- describe the procedure of testing hypothesis with an illustration.

4.1 INTRODUCTION

You may be aware that sample survey has many advantages (lower cost, lesser time, limited resources) over a population census. Therefore, we study a sample instead of the universe. We find out the sample characteristics and generalise the sample results to the population. This process is called estimation of parameters. Thus, estimation is the process of recovering the unknown parameters that were used to generate the sample data. You should have a thorough knowledge of inferential statistics, particularly estimation and hypothesis testing given in Unit 30 and Unit 31 of the course MEC 203: Quantitative Methods.

The relationship between variables in a population is given by the population regression function (PRF). The same relationship in a sample is given by the sample regression function (SRF). In simple linear regression, we estimate the population regression function (PRF) from the sample regression function (SRF). There are three methods of estimation: (i) least squares method, (ii) maximum likelihood method, and (iii) method of moments.

4.2 ESTIMATION OF BIVARIATE MODELS

A simple bivariate linear regression model is of the form:

$$Y_i = \beta_1 + \beta_2 X_i + u_i \quad \dots (4.1)$$

Here, Y_i is the dependent variable, X_i is the independent variable and u_i is the error term representing all factors other than X_i affecting Y_i . Here, β_1 and β_2 are the intercept and slope parameters respectively. For instance, Y_i may be the annual savings of the family and X_i their annual income. Let us say that the data is collected for 20 households (i.e., $n = 20$) and is used for estimation of the intercept and the slope coefficient of savings on income. Suppose we get the result as: Savings = 34560 + 0.58 Income. This tells us that each increase of Rs. 1000 in the annual income will increase the annual savings by Rs. 580.

Equation (4.1) represents the PRF which is not directly observable. We therefore estimate it from the SRF.

The estimated regression line is

$$\hat{Y}_i = \hat{\beta}_1 + \hat{\beta}_2 X_i \quad \dots (4.2)$$

In equation (4.2), note that $\hat{Y}_i$ is the expected value of the dependent variable when the independent variable takes the value X_i .

By comparing (4.1) and (4.2), we find that

$$Y_i = \hat{\beta}_1 + \hat{\beta}_2 X_i + \hat{u}_i = \hat{Y}_i + \hat{u}_i$$

$$\therefore \hat{u}_i = Y_i - \hat{Y}_i \quad \dots (4.3)$$

Remember that $\hat{u}_i$ is the residual between the actual and the estimated value of Y . The value of the error term for a particular combination (Y_i, X_i) is not known to us. We get an estimate of the error value, that is called the 'residual'. Thus, there is a difference between the error term (u_i) and the residual ($\hat{u}_i$).

4.2.1 Classical Assumptions

For carrying out the estimation of the parameters, we make the following four assumptions about the stochastic error term.

$$(i) \quad E(u_i) = 0 \quad \dots (4.4)$$

While the error is positive for some observations, it is negative for other observations. It implies that on average, the error is zero. In other words, the sum of errors across the population is zero.

$$(ii) \quad E(u_i^2) = \sigma^2 \quad \dots (4.5)$$

The variance of the error variable is σ^2 . It is constant across the population. For example, the variance of household consumption expenditure is the same for rich households as well as poor households.

$$(iii) \quad E(u_i u_j) = 0 \text{ for } i \neq j \quad \dots (4.6)$$

There is no correlation between the error terms for different observations in a dataset. In other words, the error terms of two observations are independent.

$$(iv) \quad E(X_i u_i) = 0 \quad \dots (4.7)$$

The explanatory variable is not correlated with the error term. This assumption is required for fulfilment of the properties of the estimators ($\hat{\beta}_0, \hat{\beta}_1$) on unbiasedness and consistency. In order to fulfil this condition, we assume that the explanatory variable (regressor) is exogenously given. We often keep the values of X_i fixed in repeated samples.

$$(v) \quad u_i \sim N(0, \sigma^2) \quad \dots (4.8)$$

The error variable u_i follows normal distribution with zero mean and σ standard deviation. For a given value of X , there is a whole range of values of u and a particular value is selected randomly.

The above assumptions are called the classical assumptions. These assumptions indicate the ideal conditions regarding the error variable. In empirical studies, some of the above assumptions may not be fulfilled. As we will see the fifth assumption is not required for the OLS estimation, although it is required for

hypothesis testing. In the case of maximum likelihood estimation, it is necessary to state the probability distribution function of the variables under consideration.

4.2.2 Ordinary Least Squares (OLS) Method

Our objective is to minimise the value of $\hat{u}_i$. For this purpose, we choose those values of $\hat{\beta}_1$ and $\hat{\beta}_2$, for which $\sum \hat{u}_i^2$ is the minimum. We use the term ‘least squares’ because it involves ‘the smallest sum of the squares of the errors’. From Block 3 of the course MEC 203: Quantitative Methods we know the conditions for minimization of a function: (i) first derivative should be equal to zero, and (ii) second derivative should be positive.

Let us take the square of $\hat{u}_i$ and sum it over the sample. From equation (4.3), we obtain

$$\sum \hat{u}_i^2 = \sum (Y_i - \hat{Y}_i)^2 = \sum (Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i)^2$$

As mentioned earlier, we choose the optimal value of $\hat{\beta}_1$ and $\hat{\beta}_2$. Thus, we should take the partial derivatives of $\sum \hat{u}_i^2$ with respect to $\hat{\beta}_1$ and $\hat{\beta}_2$. The term $\sum \hat{u}_i^2$ is called the Residual Sum of Squares (RSS).

The method described below is called ‘ordinary’ least squares (OLS) as it is the simplest of all – it fulfils the classical assumptions regarding the stochastic error term. In Block 3 we will introduce complexities to the OLS method (for example, weighted least squares, two-stage least squares, etc.).

By taking partial derivative of $\sum \hat{u}_i^2$ with respect to $\hat{\beta}_1$, we obtain

$$\begin{aligned} \frac{\partial \sum \hat{u}_i^2}{\partial \hat{\beta}_1} &= \frac{\partial \sum (Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i)^2}{\partial \hat{\beta}_1} \\ &= 2 \cdot (-1) \cdot \sum (Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i) \end{aligned} \quad \dots (4.9)$$

By equating (4.9) to zero, and re-arranging the terms, we obtain

$$\sum (Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i) = 0$$

or,

$$\sum Y_i = n\hat{\beta}_1 + \hat{\beta}_2 \sum X_i \quad \dots (4.10)$$

By taking partial derivative of $\sum \hat{u}_i^2$ with respect to $\hat{\beta}_2$, we obtain

$$\begin{aligned} \frac{\partial \sum \hat{u}_i^2}{\partial \hat{\beta}_2} &= \frac{\partial \sum (Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i)^2}{\partial \hat{\beta}_2} \\ &= 2 \cdot (-1) \cdot \sum (Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i) \cdot X_i \\ &= 2 \cdot (-1) \cdot \sum (X_i Y_i - \hat{\beta}_1 \cdot X_i - \hat{\beta}_2 X_i^2) \end{aligned} \quad \dots (4.11)$$

By equating (4.11) to zero, and re-arranging the terms, we obtain

$$\sum X_i Y_i = \hat{\beta}_1 \sum X_i + \hat{\beta}_2 \sum X_i^2 \quad \dots (4.12)$$

These two equations (4.10) and (4.12) are called the *normal equations* of least squares method. These are two simultaneous linear equations in two unknowns. These can be solved to obtain the values of $\hat{\beta}_1$ and $\hat{\beta}_2$.

By dividing equation (4.10) by the sample size, n , we obtain

$$\bar{Y} = \widehat{\beta}_1 + \widehat{\beta}_2 \bar{X}$$

Therefore,

$$\widehat{\beta}_1 = \bar{Y} - \widehat{\beta}_2 \bar{X} \quad \dots (4.13)$$

If we know the value of $\widehat{\beta}_2$, we can obtain the value of $\widehat{\beta}_1$ from (4.13). After rearrangement of terms in equations (4.10) and (4.11), we obtain

$$\widehat{\beta}_2 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

If we take $x_i = (X_i - \bar{X})$ and $y_i = (Y_i - \bar{Y})$, then

$$\widehat{\beta}_2 = \frac{\sum x_i y_i}{\sum x_i^2} \quad \dots (4.14)$$

4.2.3 Method of Moments

In the method of moments, we equate the sample moments with the population moments. The n^{th} moment of a random variable is the expected value of the n^{th} power. Thus, the first moment of a random variable is its arithmetic mean.

In Sub-Section 4.2.1, while mentioning the classical assumptions of the error term, we consider two population moments. Look at assumptions (4.3) and (4.6). These assumptions are

$$E(u_i) = 0 \quad \dots (4.3)$$

and,

$$E(X_i u_i) = 0 \quad \dots (4.6)$$

Using (4.1), equations (4.3) and (4.6) can be written as:

$$E(Y - \beta_1 - \beta_2 X) = 0 \quad \dots (4.15)$$

and

$$E[X(Y - \beta_1 - \beta_2 X)] = 0 \quad \dots (4.16)$$

Equations (4.15) and (4.16) can be used to obtain the estimators of β_1 and β_2 . Summing up over all the observations of the sample, and dividing by the sample size n , equations (4.15) and (4.16) become:

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \widehat{\beta}_1 - \widehat{\beta}_2 X_i) = 0 \quad \dots (4.17)$$

$$\frac{1}{n} \sum_{i=1}^n X_i (Y_i - \widehat{\beta}_1 - \widehat{\beta}_2 X_i) = 0 \quad \dots (4.18)$$

By re-arranging terms, we can write equation (4.17) as:

$$\bar{Y} = \widehat{\beta}_1 + \widehat{\beta}_2 \bar{X} \quad \dots (4.19)$$

where $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$.

Note that $\bar{Y}$ and $\bar{X}$ in (4.19) are the sample means of Y and X respectively. We therefore have

$$\widehat{\beta}_1 = \bar{Y} - \widehat{\beta}_2 \bar{X} \quad \dots (4.20)$$

Thus, if we have the slope estimator $\widehat{\beta}_2$, we can obtain the intercept estimator $\widehat{\beta}_1$ from equation (4.20) by using the sample values of $\bar{Y}$ and $\bar{X}$. The estimator of the slope coefficient $\widehat{\beta}_2$ can be obtained as follows:

Let us re-write (4.18) [by dropping $\frac{1}{n}$ and substituting for $\widehat{\beta}_1$ from (4.15)] as:

$$\sum_{i=1}^n X_i(Y_i - (\bar{Y} - \widehat{\beta}_2\bar{X}) - \widehat{\beta}_2 X_i) = 0 \quad \dots (4.21)$$

Rearranging (4.21) we get:

$$\sum_{i=1}^n X_i(Y_i - \bar{Y}) = \widehat{\beta}_2 \sum_{i=1}^n X_i(X_i - \bar{X}) \quad \dots (4.22)$$

Note that by the properties of summation, we have:

$$\sum_{i=1}^n X_i(X_i - \bar{X}) = \sum_{i=1}^n (X_i - \bar{X})^2$$

and

$$\sum_{i=1}^n X_i(Y_i - \bar{Y}) = \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

Hence, by assuming that

$$\sum_{i=1}^n (X_i - \bar{X})^2 > 0 \quad \dots (4.23)$$

An estimator of slope coefficient can be obtained as

$$\widehat{\beta}_2 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} \quad \dots (4.24)$$

Or,

$$\widehat{\beta}_2 = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2} \quad \dots (4.25)$$

The estimators of $\widehat{\beta}_1$ and $\widehat{\beta}_2$ in (4.20) and (4.25) respectively are the same as the Ordinary Least Squares (OLS) estimators given at (4.13) and (4.14). From (4.25) we find that the slope estimator $\widehat{\beta}_2$ is simply the sample covariance between X and Y divided by the sample variance of X . So, the only condition required for OLS estimators to exist is that $\sum_{i=1}^n (X_i - \bar{X})^2 > 0$. This condition is always fulfilled, except for the situation where the value of all X_i is the same. Such cases are not common. *Thus, to obtain estimates of the parameters, we should have different values of X .* It is an essential condition in econometric estimation of regression models that $n > k$.

4.2.4 Maximum Likelihood Method

You are familiar with the concept of probability density function such as normal distribution. If we have two random variables (say, X_1 and X_2), then we have a joint probability density function. The likelihood function is identical in functional form with the joint density function. The joint density function is ‘a function of the random variable outcomes’, given the value of the parameters. The interpretation of the likelihood function is the opposite of the joint density function. The likelihood function is ‘a function of the values of the parameters’, given the values of the random variable outcomes.

$$f(X|\theta) = L(\theta|X) \quad \dots (4.26)$$

where $L(\theta|X)$ is the likelihood function.

Let me explain the concept of likelihood function in terms of the error variable u_i . In Sub-Section 4.2.1, we mentioned the assumptions of the error variable u_i . As per assumption (4.8), the stochastic error variable u_i follows normal distribution with mean zero and standard deviation σ such that $u_i \sim N(0, \sigma^2)$.

Suppose we have to select the first observation of the sample. The error term u_1 could assume any random value and its probability density function is given by

$$f(u_1|\theta) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{u_1 - E(u_1)}{\sigma}\right]^2}$$

Since $E(u_1) = 0$, we have

$$f(u_1) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}u_1^2} \quad \dots (4.27)$$

Similarly, the probability density function of the second error term u_2 is given by

$$f(u_2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{u_2 - E(u_2)}{\sigma}\right]^2} = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}u_2^2}$$

As per assumption (iii), the error terms are independent of one another. We can visualize the occurrence of the stochastic error terms (u_i, u_j) as independent events. Therefore, the joint density function of u_1, u_2 is the multiplication of $P(u_1)$ and $P(u_2)$ given above. If we have data on n sample observations, we have $u_1, u_2, u_3, \dots, u_n$. The likelihood of these independent random variables (that is, u_i) is given by

$$L(\theta|u) = \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{u_i - E(u_i)}{\sigma}\right]^2} \quad \dots (4.28)$$

From (4.3) we know that $\hat{u}_i = (Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i)$.

Therefore,

$$L(\theta|u) = \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{Y_i - \beta_1 - \beta_2 X_i}{\sigma}\right]^2} \quad \dots (4.29)$$

If we take natural log of (4.28) we obtain

$$\ln L(\cdot) = -\frac{n}{2} \ln \sigma^2 - \frac{n}{2} \ln(2\pi) - \frac{1}{2} \sum \frac{(Y_i - \beta_1 - \beta_2 X_i)^2}{\sigma^2} \quad \dots (4.30)$$

There are three parameters, viz., β_1, β_2 and σ in (4.30), for which we must obtain estimators. For maximization of the likelihood function, L , we take partial derivatives of (4.32) with respect to $\tilde{\beta}_1, \tilde{\beta}_2$ and $\tilde{\sigma}$.

$$\frac{\partial \ln L}{\partial \tilde{\beta}_1} = \frac{1}{\tilde{\sigma}^2} \sum (Y_i - \tilde{\beta}_1 - \tilde{\beta}_2 X_i)(-1) \quad \dots (4.31)$$

$$\frac{\partial \ln L}{\partial \tilde{\beta}_2} = \frac{1}{\tilde{\sigma}^2} \sum (Y_i - \tilde{\beta}_1 - \tilde{\beta}_2 X_i)(-X_i) \quad \dots (4.32)$$

Setting these equations to zero (i.e., applying the first-order condition for optimization), and re-arranging terms, and denoting by $\tilde{\beta}_1, \tilde{\beta}_2$ and $\tilde{\sigma}^2$ as the 'maximum likelihood estimates (MLEs)', we get:

$$\sum Y_i = n\tilde{\beta}_1 + \tilde{\beta}_2 \sum X_i \quad \dots (4.33)$$

$$\sum Y_i X_i = \tilde{\beta}_1 \sum X_i + \tilde{\beta}_2 \sum X_i^2 \quad \dots (4.34)$$

Equations (4.35) and (4.36) are the same as the normal equations of the OLS method of estimation given at (4.10) and (4.12). Therefore, the maximum likelihood estimators of β_1, β_2 are the same as the OLS estimators.

The maximum likelihood method provides an estimator for the error variance σ^2 also. For taking partial derivative with respect to $\tilde{\sigma}$, let us re-arrange the terms in (4.30) and obtain

$$\begin{aligned} \ln L(\cdot) &= -n \cdot \frac{1}{2} \cdot \ln \sigma^2 - \frac{n}{2} \ln(2\pi) - \frac{1}{2} \sum \frac{(Y_i - \beta_1 - \beta_2 X_i)^2}{\sigma^2} \\ &= -n \cdot \ln \sigma - \frac{n}{2} \ln(2\pi) - \frac{1}{2} \cdot \sigma^{-2} \cdot RSS \\ &[\text{since } \sum (Y_i - \beta_1 - \beta_2 X_i)^2 = \sum \hat{u}_i^2 = RSS] \end{aligned}$$

Thus,

$$\begin{aligned} \frac{\partial \ln L}{\partial \tilde{\sigma}} &= -n \cdot \frac{1}{\tilde{\sigma}} - 0 - \frac{1}{2} \cdot (-2) \cdot \tilde{\sigma}^{-3} \cdot RSS \\ &= -\frac{n}{\tilde{\sigma}} + \frac{RSS}{\tilde{\sigma}^3} \end{aligned}$$

Equating the above to zero yields

$$-\frac{n}{\tilde{\sigma}} + \frac{RSS}{\tilde{\sigma}^3} = 0$$

Or,

$$\frac{RSS}{\tilde{\sigma}^2} = n$$

Or,

$$\tilde{\sigma}^2 = \frac{RSS}{n} \quad \dots (4.35)$$

Remember that the maximum likelihood estimator of σ^2 is biased, but consistent.

4.2.5 Interpretation of OLS Estimators

There could be three types of regression equations:

$$(a) \quad Y_i = \beta_1 + \beta_2 X_i + u_i \quad \dots (4.36)$$

$$(b) \quad \ln Y_i = \beta_1 + \beta_2 X_i + u_i \quad \dots (4.37)$$

$$(c) \quad \ln Y_i = \beta_1 + \beta_2 \ln X_i + u_i \quad \dots (4.38)$$

Equation (4.38) is the usual one, which we considered in (4.1). Equation (4.37) is the semi-log equation where the dependent variable is in log, while the independent variable is in natural numbers. In equation (4.38) we have both the variables in logarithmic terms, and it is called 'double-log' or 'log-linear' model. We have discussed the specification and interpretation of these models in Sub-Section 1.5.2 of Unit 1.

Let us interpret equation (4.36). Suppose we want to find the relationship between wage (w) and years of schooling (s) of workers. For a sample of total

work force in a particular year, let Y = wage measured in INR per hour, and X = education level (expressed in terms of number of years of formal education). Suppose, for a sample of size 600, the regression model obtained is

$$\hat{w} = 0.89 + 0.63s \quad \dots (4.39)$$

where $\hat{w}$ is the predicted value of the wage. The result tells us that ‘when the years of schooling is 0, i.e., the respondent is illiterate, the predicted hourly wage is Rs. 0.89 or 89 paise. If the respondent has 8 years of schooling, the predicted hourly wage is $0.89 + 0.63(8)$ or Rs. 5.93. If the person has 12 years of schooling, his hourly wage will be Rs. 8.45. Thus, when the years of schooling increases by one year, the hourly wage increases by Rs. 0.63 or 63 paise.

Check Your Progress 1

- 1) State the assumptions of the classical regression model.

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- 2) Let the regression model be $Y_i = \beta_1 + \beta_2 X_i + u_i$, where Y is hourly wage rate (measured in rupees per hour) and X is years of schooling. Also, $x_i = (X_i - \bar{X}_i)$ and $y_i = (Y_i - \bar{Y}_i)$.

Based on the following data for a sample of 526 workers, compute the OLS estimators of the intercept β_1 and slope β_2 .

$$n = 526$$

$$\sum_{i=1}^n Y_i = 3101.35$$

$$\sum_{i=1}^n X_i^2 = 87040.0$$

$$\sum_{i=1}^n y_i^2 = 7160.414$$

$$\sum_{i=1}^n X_i = 6608.0$$

$$\sum_{i=1}^n X_i Y_i = 41140.65$$

$$\sum_{i=1}^n x_i^2 = 4025.43$$

$$\sum_{i=1}^n Y_i^2 = 25446.29$$

$$\sum_{i=1}^n x_i y_i = 2179.204$$

$$\sum_{i=1}^n \hat{u}_i^2 = 5980.682$$

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- 3) Interpret the slope coefficient estimated in question (2) above.

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- 4) Specify the likelihood function for the simple regression model.

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4.3 STANDARD ERROR OF THE ESTIMATORS

Recall that the concepts ‘estimator’ and ‘estimate’ are different. While the term estimator relates to the formula or the rule, the term estimate reflects the numerical value. By applying the sample formula, we may obtain different values of $\hat{\beta}_1$ and $\hat{\beta}_2$, as we take different samples. For example, the estimator of $\hat{\beta}_2$ is given by $\hat{\beta}_2 = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$. If we take two different samples, and apply the same estimator (formula), we will get two different estimates (values) of $\hat{\beta}_2$. For hypothesis testing, we need the standard error of the estimators $\hat{\beta}_1$ and $\hat{\beta}_2$.

4.3.1 Standard Error of the OLS Estimators

We need a measure to know the reliability (or precision) of the estimators $\hat{\beta}_1$ and $\hat{\beta}_2$. We need this because the least-square estimators are functions of the sample data and would change with sample. Precision is measured by the standard error (*se*) of the estimators. With the classical assumptions, the variance of the estimators $\hat{\beta}_1$ and $\hat{\beta}_2$ are given below. The standard error of an estimator is the positive square root of its variance.

$$var(\hat{\beta}_2) = \frac{\sigma^2}{\sum x_i^2} \quad \dots (4.40)$$

$$se(\hat{\beta}_2) = \frac{\sigma}{\sqrt{\sum x_i^2}} \quad \dots (4.41)$$

$$var(\hat{\beta}_1) = \frac{\sum x_i^2}{n \sum x_i^2} \sigma^2 \quad \dots (4.42)$$

$$se(\hat{\beta}_1) = \sqrt{\frac{\sum x_i^2}{n \sum x_i^2}} \sigma \quad \dots (4.43)$$

We observe that

- a) The variance of $\widehat{\beta}_2$ is directly proportional to σ^2 and inversely proportional to $\sum x_i^2$. Thus, if the variation in X values increases, the value of $\sum x_i^2$ will increase. Therefore, the X value should be widely dispersed. In other words, β_2 can be measured with greater accuracy if there is considerable variation in X values.
- b) As sample size n increases, the value of $\sum x_i^2$ will increase. Therefore, it is useful to have a large sample.
- c) The variance of $\widehat{\beta}_1$ is directly proportional to σ^2 and $\sum X_i^2$. However, it is inversely proportional to $\sum x_i^2$ and the sample size n .

The estimators $\widehat{\beta}_1$ and $\widehat{\beta}_2$ will not just vary from one sample to another. The estimates $\widehat{\beta}_1$ and $\widehat{\beta}_2$ be dependent on each other in a sample. Such dependency is measured by the covariance between them.

4.3.2 Estimator of Error Variance

The variance of $\widehat{\beta}_1$ and $\widehat{\beta}_2$ depends on σ^2 , the variance of u_i . The error variance (that is, the variance of u_i), however, is not known to us. The estimator of σ^2 can be obtained by OLS method and the maximum likelihood method. The OLS estimator $\widehat{\sigma}^2$ is given by

$$\widehat{\sigma}^2 = \frac{\sum \widehat{u}_i^2}{n-2} = \frac{RSS}{n-2} \quad \dots (4.44)$$

Note that $\widehat{\sigma}^2$ is the OLS estimator of the true but unknown σ^2 with $(n - 2)$ degrees of freedom. Once $\sum \widehat{u}_i^2$ is estimated as the sum of squares of the residuals (RSS) given at equation (4.3), we can calculate $\widehat{\sigma}^2$ by using (4.44).

In the maximum likelihood (ML) method we obtained the estimator $\tilde{\sigma}^2 = \frac{RSS}{n}$.

Thus, the ML estimator of σ^2 is biased. However, it is consistent, in the sense that as we increase the sample size infinitely, $\tilde{\sigma}^2$ approaches the true value of the parameter σ^2 .

4.4 PROPERTIES OF THE OLS ESTIMATORS

The properties of OLS estimators can be classified as: (i) algebraic or numerical, and (ii) statistical. The statistical properties pertain to unbiasedness, consistency and efficiency of the OLS estimators. The algebraic properties of OLS estimators pertain to the relationship among the variables in the regression model.

4.4.1 Algebraic Properties

The algebraic properties of OLS estimators are as follows:

- (i) The OLS estimators provide point estimates. Thus, *each estimator provides a single value of the population parameter.*
- (ii) The sample regression line passes through the sample means of Y and X . The estimated regression line is given by $\hat{Y}_i = \hat{\beta}_1 + \hat{\beta}_2 X_i$. If we add up over the sample, and divide it by n , we obtain $\bar{Y} = \hat{\beta}_1 + \hat{\beta}_2 \bar{X}$. Thus, when X takes the value $\bar{X}$, the dependent variable Y takes the value $\bar{Y}$. The regression line passes through the point $(\bar{X}, \bar{Y})$

(iii) For the dependent variable, the mean of actual the actual value (Y) is equal to the mean of the estimated value ($\hat{Y}$). If we substitute the value of $\hat{\beta}_1$ from equation (4.13) in (4.2), we obtain

$$\begin{aligned}\hat{Y}_i &= \hat{\beta}_1 + \hat{\beta}_2 X_i \\ &= (\bar{Y} - \hat{\beta}_2 \bar{X}) + \hat{\beta}_2 X_i = \bar{Y} - \hat{\beta}_2 (X_i - \bar{X})\end{aligned}$$

Let us add up both the sides of equation (4.38), and divide by the sample size n . Further, let us apply the fact that $\sum (X_i - \bar{X}) = 0$. This gives us

$$\bar{\hat{Y}} = \bar{Y} \quad \dots (4.45)$$

- (iii) The sample regression function can be expressed in deviation form as $\hat{y}_i = \hat{\beta}_2 x_i$.

We know that $\sum \hat{u}_i = 0$.

$$\hat{Y}_i = \hat{\beta}_1 + \hat{\beta}_2 X_i \quad \dots (4.2)$$

and

$$Y_i = (\hat{Y}_i + \hat{u}_i) = (\hat{\beta}_1 + \hat{\beta}_2 X_i) + \hat{u}_i$$

Or,

$$Y_i = \hat{\beta}_1 + \hat{\beta}_2 X_i + \hat{u}_i \quad \dots (4.3)$$

Summing up both the sides of equation (4.2) and dividing by the sample size n , we obtain

$$\bar{\hat{Y}} = \hat{\beta}_1 + \hat{\beta}_2 \bar{X}$$

Since $\bar{\hat{Y}} = \bar{Y}$, we have

$$\bar{Y} = \hat{\beta}_1 + \hat{\beta}_2 \bar{X} \quad \dots (4.46)$$

Subtracting (4.43) from (4.3) we get

$$(Y_i - \bar{Y}) = \hat{\beta}_2 (X_i - \bar{X}) + \hat{u}_i \quad \dots (4.47)$$

If we define y_i and x_i as deviations from their sample means, the above can be expressed as

$$y_i = \hat{\beta}_2 x_i + \hat{u}_i \quad \dots (4.48)$$

From (4.2) above, we have

$$\bar{Y} = \widehat{\beta}_1 + \widehat{\beta}_2 \bar{X}$$

Subtracting the above from (4.2), we obtain the regression line in deviation form as

$$\widehat{y}_i = \widehat{\beta}_2 x_i \quad \dots (4.49)$$

- (iv) The predicted value $\widehat{y}_i$ and the residual $\widehat{u}_i$ are not correlated. It implies, we have to prove that $E(\widehat{y}_i \widehat{u}_i) = 0$, or $\frac{1}{n} \sum \widehat{y}_i \widehat{u}_i = 0$.

From (4.44) $\widehat{y}_i = \widehat{\beta}_2 x_i$. If we multiply both the sides by $\widehat{u}_i$ and take summation over all the observations, we get:

$$\begin{aligned} \sum \widehat{y}_i \widehat{u}_i &= \widehat{\beta}_2 \sum x_i \widehat{u}_i \\ &= \widehat{\beta}_2 \sum x_i (y_i - \widehat{\beta}_2 x_i) \\ &= \widehat{\beta}_2 \sum x_i y_i - \widehat{\beta}_2^2 \sum x_i^2 \\ &= \widehat{\beta}_2 \cdot \frac{\widehat{\beta}_2}{\widehat{\beta}_2} \sum x_i y_i - \widehat{\beta}_2^2 \sum x_i^2 \\ &= \widehat{\beta}_2 \cdot \frac{\widehat{\beta}_2}{\widehat{\beta}_2} \sum x_i y_i - \widehat{\beta}_2^2 \sum x_i^2 \\ &= \widehat{\beta}_2^2 \cdot \frac{\sum x_i^2}{\sum x_i y_i} \cdot \sum x_i y_i - \widehat{\beta}_2^2 \sum x_i^2 = 0 \\ &= \widehat{\beta}_2^2 \sum x_i^2 - \widehat{\beta}_2^2 \sum x_i^2 = 0 \quad \dots (4.50) \end{aligned}$$

Hence, the residuals $\widehat{u}_i$ and Y_i are not correlated.

- (v) The residual $\widehat{u}_i$ and X_i are not correlated. This implies, $E(X_i \widehat{u}_i) = 0$. In other words, $\frac{1}{n} \sum X_i \widehat{u}_i = 0$.

While taking the partial derivative of $\sum \widehat{u}_i^2$ with respect to $\widehat{\beta}_2$ (see equation (4.11)) we find that

$$\begin{aligned} \frac{\partial \sum \widehat{u}_i^2}{\partial \widehat{\beta}_2} &= \frac{\partial \sum (Y_i - \widehat{\beta}_1 - \widehat{\beta}_2 X_i)^2}{\partial \widehat{\beta}_2} = 0 \\ \text{Or, } 2 \cdot (-1) \cdot \sum (Y_i - \widehat{\beta}_1 - \widehat{\beta}_2 X_i) \cdot X_i &= 0 \\ \text{Or, } \sum X_i (Y_i - \widehat{\beta}_1 - \widehat{\beta}_2 X_i) &= 0 \\ \text{Or, } \sum X_i \widehat{u}_i &= 0 \quad \dots (4.51) \end{aligned}$$

4.4.2 Statistical Properties

The OLS estimators fulfil certain statistical or theoretical properties. We specify them below.

(i) Unbiasedness

The OLS estimators are unbiased.

$$E(\widehat{\beta}_2) = E\left(\frac{\sum x_i y_i}{\sum x_i^2}\right) = E\left(\frac{\sum x_i (\beta x_i + u_i)}{\sum x_i^2}\right) = E\left(\frac{\sum \beta x_i^2 + \sum x_i u_i}{\sum x_i^2}\right)$$

$$= \beta + E\left(\frac{\sum x_i u_i}{\sum x_i^2}\right) = \beta + \left(\frac{\sum x_i E(u_i)}{\sum x_i^2}\right) = \beta + 0 = \beta$$

$$\text{Thus, } E(\widehat{\beta}_2) = \beta_2 \quad \dots (4.52)$$

(ii) Consistency

The OLS estimators are consistent. As we increase the sample size, $\widehat{\beta}_2$ converges to β_2 .

$$\widehat{\beta}_2 = \frac{\sum x_i y_i}{\sum x_i^2} = \beta + \frac{\sum x_i u_i}{\sum x_i^2} = \beta + \frac{\sum x_i u_i / n}{\sum x_i^2 / n} \quad \dots (4.53)$$

As $n \rightarrow \infty$, we observe that the covariance between X and u is zero, that is, $\sum x_i u_i / n \rightarrow 0$. Therefore, $\widehat{\beta}_2$ converges to β_2 .

(iii) Best Linear Unbiased Estimator (BLUE)

The OLS estimators are the best linear unbiased estimators (BLUE). This property is also called the ‘**Gauss-Markov theorem**’. The theorem states that ‘given the assumptions of classical linear regression models (CLRM), the least squares estimators have the minimum variance in the class of all linear unbiased estimators’. More specifically, the OLS estimator of $\widehat{\beta}_2$ is said to be BLUE of β_2 . The OLS estimator $\widehat{\beta}_2$ has the following properties:

(i) Linear: The OLS estimator is a linear function of the dependent variable. As you know, $\widehat{\beta}_2 = \frac{\sum x_i y_i}{\sum x_i^2}$. We can present it as

$$\widehat{\beta}_2 = \sum c_i y_i \quad \dots (4.54)$$

$$\text{where } c_i = \frac{x_i}{\sum x_i^2}.$$

(ii) Unbiased: The OLS estimator is unbiased. We have proved in equation (4.52) that $\widehat{\beta}_2$ is unbiased estimator of β_2 .

(iii) Efficient: The OLS estimator is efficient in the sense that in the class of all linear unbiased estimators, it has the minimum or the least variance. In order to prove it, we proceed as follows:

Let us consider any linear unbiased estimator b_2 such that

$$b_2 = \sum d_i y_i \quad \dots (4.55)$$

$$E(b_2) = \sum d_i E(y_i) = \sum d_i E(\beta_2 x_i) = \beta_2 \sum d_i x_i$$

For b_2 to be unbiased, $E(b_2) = \beta_2$. For this, we should have

$$\sum d_i x_i = 1 \quad \dots (4.56)$$

Since $b_2 = \sum d_i y_i$, we should have

$$\text{var}(b_2) = \sum d_i^2 \sigma^2 \quad \dots (4.57)$$

Our objective is to find out the value of d_i for which $\sum d_i^2 \sigma^2$ is minimum. Also, it should fulfil the condition that $\sum d_i x_i = 1$. Since σ^2 is a constant, we need to minimize the following Lagrangian function:

$$\sum d_i^2 - \lambda(\sum d_i x_i - 1) \quad \dots (4.58)$$

where λ is the Lagrangian multiplier.

If we differentiate equation (4.58) with respect to d_i , and equate it to zero, we obtain

$$2d_i - \lambda x_i = 0$$

Or,

$$d_i = \frac{\lambda}{2} x_i \quad \dots (4.59)$$

Multiplying both sides of equation (4.59) by x_i and summing over the sample, we obtain

$$\sum d_i x_i = \frac{\lambda}{2} \sum x_i^2 \quad \dots (4.60)$$

Recall from (4.56) that $\sum d_i x_i = 1$. Hence, $\frac{\lambda}{2} \sum x_i^2 = 1$. Therefore, $\lambda = \frac{2}{\sum x_i^2}$.

Substituting this value of λ in (4.59), we obtain

$$d_i = \frac{2}{\sum x_i^2} \cdot \frac{1}{2} x_i = \frac{x_i}{\sum x_i^2} \quad \dots (4.61)$$

Therefore, the minimum variance of b_2 is achieved when (see equation (4.55)),

$$b_2 = \sum d_i y_i = \sum \frac{x_i}{\sum x_i^2} y_i = \frac{\sum x_i y_i}{\sum x_i^2} \quad \dots (4.62)$$

Compare equation (4.62) with the OLS estimator given at equation (4.14). Both are the same. Thus, the OLS estimator has the minimum variance among the linear unbiased estimators; and it is BLUE.

The Gauss-Markov theorem holds only if the assumptions of CLRM are satisfied. This ensures that we cannot get any other estimator with variance lower than that of the OLS estimator. A remarkable feature of the theorem is that it makes no assumption about the distribution of the error term u_i and Y_i . Any violation of the CLRM assumptions leads to invalidity of the theorem. For instance, if we have a regression model which is non-linear in parameters, we may obtain estimators which are better than OLS estimators. Another example is the case of violation of the homoscedastic assumption. Under such situations, the estimators may be unbiased and consistent, but the variance will not be the minimum even in the class of linear estimators. It is important to note that the statistical properties of the OLS estimator are valid, irrespective of the sample size.

Check Your Progress 2

- 1) State the variance of the OLS estimators.

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- 2) State the numerical properties of the OLS estimators.

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- 3) State the statistical properties of the OLS estimators.

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- 4) State the central features of Gauss Markov Theorem.

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- 5) Using the data given for question 2 in CYP 1, calculate the estimator of the error variance σ^2 .

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4.5 GOODNESS OF FIT

Let us now examine how we can ascertain the ‘goodness of fit’ of the estimated regression line. A perfect fit would be obtained if all the observations were on the regression line. But this is a rare possibility. We normally get a distribution in such a way that some observations are below, and some observations are above the regression line. The distance between the actual value Y from the expected value $\hat{Y}$ will have both positive and negative values. Thus, the residual ($\hat{u}_i$) can

be either positive or negative. What we require is that the value of this residual $\hat{u}_i$ is as small as possible. A measure of how well the sample regression line fits the data is given by the ‘coefficient of determination’ (r^2). Note that the notation r^2 is usually used for bi-variate regressions, while R^2 is used for multiple regression. How do we compute r^2 ? Let us once again consider our SRF as in equation (4.3), i.e., $Y_i = \hat{Y}_i + \hat{u}_i$. In deviation form, this can be written as:

$$y_i = \hat{y}_i + \hat{u}_i \quad \dots (4.63)$$

Squaring (4.30) on both sides and summing up we get:

$$\begin{aligned} \sum y_i^2 &= \sum \hat{y}_i^2 + \sum \hat{u}_i^2 + 2 \sum \hat{y}_i \hat{u}_i \\ &= \sum \hat{y}_i^2 + \sum \hat{u}_i^2 \quad (\text{since } \sum \hat{y}_i \hat{u}_i = 0) \\ &= \hat{\beta}_2^2 \sum x_i^2 + \sum \hat{u}_i^2 \quad \dots (4.64) \end{aligned}$$

Equation (4.64) contains the following sum of squares:

- (i) $\sum y_i^2$ represents the total variation of Y values about the mean. This is termed as the ‘total sum of squares (TSS)’.
- (ii) $\hat{\beta}_2^2 \sum x_i^2$ represents the variation of estimated Y values about their mean. This is termed as the ‘explained sum of squares (ESS)’ or variation due to regression, or due to the explanatory variables.
- (iii) $\sum \hat{u}_i^2$ represents the residual or the unexplained variation of Y and is termed as the ‘residual sum of squares (RSS)’.

We can therefore write (4.64) in terms of the sum of the squares as:

$$TSS = ESS + RSS \quad \dots (4.65)$$

The total variations in Y (about its mean) is thus segregated into two parts: one part of the variation attributed to the regression and the second part to the random error. Dividing (4.65) by TSS on both sides we get:

$$1 = \frac{ESS}{TSS} + \frac{RSS}{TSS} \quad \dots (4.66)$$

$$1 = \frac{\sum (\hat{Y}_i - \bar{Y})^2}{\sum (Y_i - \bar{Y})^2} + \frac{\sum \hat{u}_i^2}{\sum (Y_i - \bar{Y})^2} \quad \dots (4.67)$$

From (4.63), r^2 can be defined as:

$$r^2 = \frac{\sum (\hat{Y}_i - \bar{Y})^2}{\sum (Y_i - \bar{Y})^2} = \frac{ESS}{TSS} \quad \dots (4.68)$$

Therefore, equation (4.67) can also be written as:

$$\begin{aligned} r^2 &= 1 - \frac{\sum \hat{u}_i^2}{\sum (Y_i - \bar{Y})^2} \\ &= 1 - \frac{RSS}{TSS} \quad \dots (4.69) \end{aligned}$$

$$\text{Alternatively, } r^2 = \frac{ESS}{TSS} = \frac{\sum (\hat{Y}_i)^2}{\sum y_i^2} = \frac{\hat{\beta}_2^2 \sum x_i^2}{\sum y_i^2}$$

$$= \widehat{\beta}_2^2 \left(\frac{\sum x_i^2}{\sum y_i^2} \right) \quad \dots (4.70)$$

Dividing both the numerator and the denominator of the RHS of (4.70) by the sample size n we get:

$$r^2 = \widehat{\beta}_2^2 \left(\frac{S_x^2}{S_y^2} \right) \quad \dots (4.71)$$

where S_x^2 and S_y^2 are the sample variances of X and Y respectively. Substituting for $\widehat{\beta}_2$ from (4.14), equation (4.70) can be written as:

$$r^2 = \frac{(\sum x_i y_i)^2}{\sum x_i^2 \sum y_i^2} \quad \dots (4.72)$$

Therefore, as a formal definition of r^2 , we can say that r^2 is the measure of the proportion of total variation in Y that is explained by X in the regression model. It provides the magnitude or the extent to which variation in the independent variable can affect the variation in the dependent variable. The value of r^2 is always non-negative and ranges between 0 to 1, that is, $0 \leq r^2 \leq 1$. A value of 1 would mean a perfect fit, i.e., $\widehat{Y}_i = Y_i$, whereas $r^2 = 0$ means that there is absolutely no relationship between the dependent and the explanatory variable.

4.6 TESTING OF HYPOTHESIS

The main idea of hypothesis testing is to examine if the sample value is compatible (i.e., sufficiently close) to the population value. For example, let us say that the theory or previous experience says that the true slope coefficient of β_2 is zero. But from the sample, we obtain the estimate as $\widehat{\beta}_2 = 0.674$. Now, we need to test whether the estimate obtained is consistent with the hypothesis. In case it is consistent, we do not reject the hypothesis. Here, the hypothesis that β_2 is zero is called as the ‘null hypothesis or H_0 ’. This is compared against the alternative hypothesis H_1 that β_2 is different from zero. The alternative hypothesis may be simple like $H_1: \beta_2 \neq 0$.

Hypothesis testing requires developing rules or procedures to decide whether to reject or not-reject the null hypothesis. There are two common approaches to hypothesis testing: (i) confidence interval approach, and (ii) test-of-significance approach. Both the approaches are based on the fact that the variable of our concern has some probability distribution. Hypothesis testing involves making statements or claims about the value of the parameter given such a distribution. Here we first focus on the ‘test-of-significance approach’.

The test-of-significance approach refers to a procedure for testing a null hypothesis (H_0) using the sample results. The procedure involves obtaining the test statistic (estimator) and the distribution of such a statistic under the null hypothesis. Based on the value of the test statistic thus obtained, the decision to accept or reject the null hypothesis is made. Under the assumption of normal distribution, we can write the test statistic as:

$$t = \frac{\widehat{\beta}_2 - \beta_2}{se(\widehat{\beta}_2)} \quad \dots (4.73)$$

Equation (4.70) follows t -distribution with $(n - 2)$ degrees of freedom in the simple regression model. If the value of β_2 is specified under the null hypothesis, we can compute the value of t -statistic using the sample data. Since this statistic (t) follows t -distribution, its confidence interval can be stated as:

$$Pr \left[-t_{\alpha/2} \leq \frac{\widehat{\beta}_2 - \beta_2^*}{se(\widehat{\beta}_2)} \leq t_{\alpha/2} \right] = (1 - \alpha) \quad \dots (4.74)$$

where β_2^* is the value of β_2 under the null hypothesis and $-t_{\alpha/2}$ and $t_{\alpha/2}$ are the critical t -values (from the t -Table) for $\frac{\alpha}{2}$ level of significance with $(n - 2)$ degrees of freedom. Rearranging (4.74) we get:

$$Pr \left[\beta_2^* - t_{\alpha/2} se(\widehat{\beta}_2) \leq \widehat{\beta}_2 \leq \beta_2^* + t_{\alpha/2} se(\widehat{\beta}_2) \right] = (1 - \alpha) \quad \dots (4.75)$$

Equation (4.75) gives the interval in which $\widehat{\beta}_2$ will lie with a probability of $(1 - \alpha)$ if $\beta_2 = \beta_2^*$. The $100(1 - \alpha) \%$ confidence interval is termed as the ‘region of acceptance’ of the null hypothesis. The region beyond this is termed as the ‘region of rejection’ or the ‘critical region’. In the confidence interval approach, a true value of β_2 is hypothesized and then it is examined if the estimated $\widehat{\beta}_2$ lies within the region of acceptance or not.

Example 4.1

Let us consider an example where it is hypothesized that the true value of $\beta_2 = 0.3$, i.e., $H_0: \beta_2 = 0.3$. The H_0 is tested against the alternative hypothesis $H_1: \beta_2 \neq 0.3$. Let $\widehat{\beta}_2 = 0.5091$ and $se(\widehat{\beta}_2) = 0.0357$ with $df = 8$. Let us assume that $\alpha = 5 \%$ so that $t_{\alpha/2} = 2.306$. Now using (4.75) we have the confidence interval for $\widehat{\beta}_2$ as:

$$Pr[0.2177 \leq \widehat{\beta}_2 \leq 0.3823] = 0.95 \quad \dots (4.76)$$

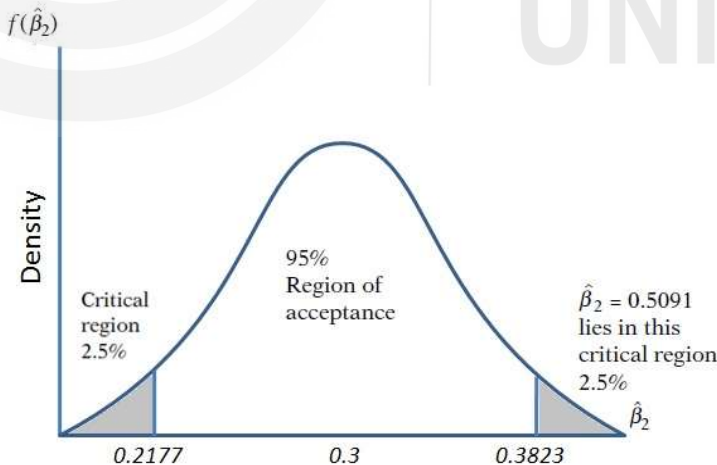


Fig.4.1: The 95% Confidence Interval for $\widehat{\beta}_2$ (under $H_0: \beta_2 = 0.3$)

Equation (4.76) is represented in Fig. 4.1. As the estimated $\widehat{\beta}_2$ lies in the critical region, we reject the null $H_0: \beta_2 = 0.3$. This can also be verified empirically by calculating the t -value using (4.73) as:

$$t = \frac{0.5091 - 0.3}{0.0357} = 5.86$$

This is again greater than 2.306, leading to the rejection of $H_0: \beta_2 = 0.3$. From (4.70), it is evident that if $\hat{\beta}_2$ is equal to the hypothesized β_2 , the t -value will be 0. As the difference between the two increases, the $|t|$ will become larger. Note that the t -value can be both positive and negative. Thus, large $|t|$ is an indication of rejection of null hypothesis.

When the alternative hypothesis is like $H_1: \beta_2 \neq 0.3$, it is a case of two-tailed or two-sided test. This is because we consider the two extreme tails of the probability distribution as the region of rejection (as in Fig. 4.1). If we had the H_0 as $\beta_2 \leq 0.3$ and H_1 as $\beta_2 > 0.3$, we use one sided (or, one-tail) test (or right-tail test) as shown in Fig.4.2. The test procedure is the same as discussed before. The only difference is that in one-tailed test, the upper critical value corresponds to $t_\alpha = t_{0.05}$ or 5% level. The lower-tail is no longer considered in this case. The choice of two-tailed or one-tailed test depends on the nature of the alternative hypothesis formulated based on a-priori considerations. The decision on the type of t -test for hypothesis testing is summarized in Table 4.1.

4.7 FORECASTING

Regression is useful for forecasting the values of Y corresponding to the given X values. This may be the conditional mean value of Y for the selected value of X , say X_0 , representing a point on the regression line itself. This is called ‘mean forecasting’. Alternatively, it can be the forecasting of an individual value of Y corresponding to X_0 . This is called ‘individual forecasting’.

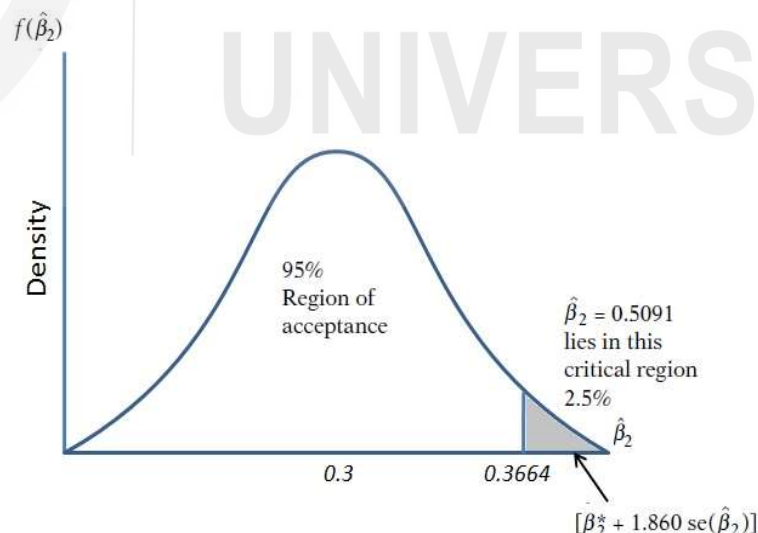


Fig.4.2: One-Tailed Test of Significance.

Table 4.1: Decision Rules for t -Test

Type of Hypothesis	H_0 : Null hypothesis	H_1 : Alternative Hypothesis	Decision – Reject H_0 if
Two-tail	$\beta_2 = \beta_2^*$	$\beta_2 \neq \beta_2^*$	$ t > t_{\alpha/2}, df$
Right-tail	$\beta_2 \leq \beta_2^*$	$\beta_2 > \beta_2^*$	$t > t_{\alpha}, df$
Left-tail	$\beta_2 \geq \beta_2^*$	$\beta_2 < \beta_2^*$	$t < -t_{\alpha}, df$

Notes: β_2^* is the hypothesized value of β_2 ; $|t|$ is the absolute value of t , $t_{\alpha/2}$ and t_{α} are the critical values of α at $\alpha/2$ level of significance; df is the degrees of freedom.

4.7.1 Mean Forecasting

Suppose we want to predict the conditional mean of Y given $X_o = 100$, i.e., $E(Y|X_o = 100)$. Such a point estimate of the mean prediction is given as:

$$\hat{Y}_o = \hat{\beta}_1 + \hat{\beta}_2 X_o \quad \dots (4.77)$$

Since $\hat{Y}_o$ is the estimator, the difference between this estimator and the true value indicates the predicted or the forecasted error. In this context, the variance of the estimator is given by:

$$var(\hat{Y}_o) = \sigma^2 \left[\frac{1}{n} + \frac{(X_o - \bar{X})^2}{\sum x_i^2} \right] \quad \dots (4.78)$$

We can compute the t -value by substituting the unknown σ^2 with its unbiased estimator $\hat{\sigma}^2$. Then the statistic:

t = follows t -distribution with $(n - 2)$ df . Its confidence interval is given by:

$$Pr \left[\hat{\beta}_1 + \hat{\beta}_2 X_o - t_{\alpha/2} se(\hat{Y}_o) \leq \beta_1 + \beta_2 X_o \leq \hat{\beta}_1 + \hat{\beta}_2 X_o + t_{\alpha/2} se(\hat{Y}_o) \right] = 1 - \alpha \quad \dots (4.79)$$

4.7.2 Individual Forecasting

Suppose we are interested in forecasting an individual Y value, Y_o , for the corresponding given value of X_o . Then, the variance of the best linear unbiased estimator of Y_o is given by:

$$var(Y_o - \hat{Y}_o) = E \left((Y_o - \hat{Y}_o)^2 \right) = \sigma^2 \left[1 + \frac{1}{n} + \frac{(X_o - \bar{X})^2}{\sum x_i^2} \right] \quad \dots (4.80)$$

Substituting the unknown σ^2 with the unbiased estimators $\hat{\sigma}^2$ [by using the square root of (4.80) for se of $\hat{Y}_o$], the statistic in (4.73) can be computed. Note that: (i) for each value of X plotted, we get a confidence band, and that, (ii) the 95% confidence interval of any $\hat{Y}_o$ is always wider than the confidence interval for the mean value Y_o . In other words, the width of the confidence band is smallest when $X_o = \bar{X}$ and it gets wider as X_o moves away from $\bar{X}$. This means that the predictive or forecasting capacity of an X_o decreases considerably as it

departs more and more from $\bar{X}$. We therefore need to be cautious in extrapolating the regression line for the purpose of predicting $E(Y|X_o)$.

Check Your Progress 3

- 1) Look into question 2 of CYP 1 again. Compute the value of r^2 . Also, interpret the calculated value of r^2 .

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- 2) Using the value of r^2 computed above, compute the test statistic for testing the null hypothesis $H_0: \beta_2 = 0$. Take the alternative hypothesis as $H_1: \beta_2 \neq 0$. Test at 5% significance level ($\alpha = 0.05$). Interpret the test outcome.

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- 3) While predicting the $E(Y|X_o)$, i.e., Y_o for the given X_o representing a point on the regression line, why does the predictive capacity of X_o decrease as the distance of X_o from $\bar{X}$ increases?

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4.7 LET US SUM UP

In this Unit we began with the specification of two-variable regression and its estimation methods. There are three methods of estimation: method of least squares, method of moments, and method of maximum likelihood. We specified the classical assumptions regarding the stochastic error term. The algebraic properties and the statistical properties of the regression model are spelt out. The explanatory power or goodness of fit of a regression model is given by the coefficient of determination. Subsequently, we discussed the procedure of testing a hypothesis. Many related concepts like coefficient of determination, level of significance, critical region and confidence interval are also explained in the unit.

4.8 KEYWORDS

Confidence Interval	It refers to a rule used to construct an interval. The interval keeps the estimated values over repeated samples within a margin of error limited to the level of significance (confidence level) chosen.
Confidence Level	It refers to the percentage of samples in which the estimated sample value remains closer to the population value.
Consistent Estimator	An estimator that converges in probability to the population parameter as the sample size grows without bound.
Heteroscedasticity	It refers to a situation where the variance of the error term is not constant.
OLS Method	It refers to the method of estimation in which the 'sum of squared residuals' is minimum.
R-Squared	It refers to the proportion of the total variation in the dependent variable that is explained by the independent variables included in the regression model.

4.9 SOME USEFUL BOOKS

- Gujarati, D N and D C Porter (2009), *Basic Econometrics*, Fifth Edition, McGraw Hill
- Johnston, J., Dinardo, J. (2007), "*Econometric Methods*", 4th Edition, McGraw-Hill Education (Asia), International Edition, pp. 109-125
- Maddala, G. S. and K. Lahiri (2009), *Introduction to Econometrics*, Fourth Edition, John Wiley and Sons
- Wooldridge, Jeffrey (2022), *Introductory Econometrics*, Seventh Edition, Cengage Learning

4.10 ANSWERS/ HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Go through Sub-Section 4.21 and answer.

$$2) \quad \widehat{\beta}_2 = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2} = \frac{2179.204}{4025.43} = 0.541326$$

$$\widehat{\beta}_1 = \bar{Y} - \widehat{\beta}_2 \bar{X}$$

$$\bar{Y} = \frac{\sum_{i=1}^n Y_i}{n} = \frac{3101.35}{526} = 5.896103$$

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n} = \frac{6608.0}{526} = 12.5627$$

$$\text{Therefore, } \widehat{\beta}_1 = 5.896103 - 0.541326 (12.5627) = -0.9043$$

- 3) The computed value of $\widehat{\beta}_2 = 0.541326$ means that for every one-year increase (decrease) in formal education of X_i (measured in years), the hourly wage rate Y_i (measured in dollars per hour) increases (decreases) by 0.5413 dollars per hour or 54.13 cents per hour.
- 4) You should describe the simple regression model and its residual in terms of equation (4.3) in the text. Subsequently, you should specify the normal probability density function of the error term and the likelihood function as equation (4.32).

Check Your Progress 2

- 1) The variance of the OLS estimators is given in Section 4.3.
- 2) The algebraic properties of OLS estimators are described in Sub-Section 4.4.1.
- 3) The statistical properties of OLS estimators are unbiasedness, consistency and BLUE. Go through Sub-Section 4.4.2 for details.
- 4) The Gauss-Markov theorem states that given the assumptions of CLRM, the least squares estimators have minimum variance in the class of all linear unbiased estimators, i.e., they are BLUE or 'best linear unbiased estimator'.
- 3) The estimators of the homoscedastic variance of the error term u_i is given by $\hat{\sigma}^2 = \frac{\sum \hat{u}_i^2}{n-2}$. We have $\sum_{i=1}^n \hat{u}_i^2 = 5980.682$, $n = 526$. Therefore, $\hat{\sigma}^2 = \frac{5980.682}{526-2} = \frac{5980.682}{524} = 11.4135$.

Check Your Progress 3

$$\begin{aligned} 1) \quad r^2 &= 1 - \frac{RSS}{TSS} = 1 - \frac{\sum \hat{u}_i^2}{\sum_{i=1}^n Y_i^2} \\ &= 1 - \frac{5980.682}{7160.414} = 1 - 0.8352 = 0.1648 \end{aligned}$$

This means that 16.48 per cent of the total sample variation in Y_i , the hourly wage rate, is explained by or attributable to the variation in regressor X_i , the years of formal education.

- 2) Given that $H_0: \beta_2 = 0$ and $H_1: \beta_2 \neq 0$, we need to compute the test statistic $t = \frac{\widehat{\beta}_2 - \beta_2}{se(\widehat{\beta}_2)}$. We also need to calculate the estimated standard error of $\widehat{\beta}_2$ as $\widehat{se}(\widehat{\beta}_2) = \sqrt{\widehat{var}(\widehat{\beta}_2)} = \sqrt{0.002835} = 0.0532$. Hence, $t = \frac{\widehat{\beta}_2 - \beta_2}{se(\widehat{\beta}_2)} = \frac{0.5413 - 0}{0.0532} = 10.1667$. The critical value of $t(524)$ at 5% significance level

$ist_{\alpha/2}(524) = t_{0.025}(524) = 1.960$. Thus, at 5% significance level ($\alpha = 0.05$), $|t| = 10.167 > t_{0.025}(\infty) = 1.960$. Therefore, we reject the null $H_0: \beta_2 = 0$ at 5% level of significance. This indicates the existence of a relationship between the wage rate and the number of years of formal education in our sample.

- 3) The 95% confidence interval of Y_0 for the given X_0 is wider than the mean value Y_0 . When the 95% confidence interval for each value of X is plotted, we get a confidence band. The width of the confidence band is the smallest when $X_0 = \bar{X}$ and it gets wider as X_0 moves away from $\bar{X}$ value. Thus, the predictive capacity of X_0 decreases as it departs from the $\bar{X}$.



UNIT 5 RESIDUAL ANALYSIS *

Structure

- 5.0 Objectives
- 5.1 Introduction
- 5.2 Issues in Estimation
- 5.3 Analysis of Residuals
 - 5.3.1 Residual Plot
 - 5.3.2 Detection of Potential Problems
 - 5.3.3 Residuals against predicted Values
 - 5.3.4 Residual Plot over Time
- 5.4 Outliers
- 5.5 Visual Detection of Heteroscedasticity
- 5.6 Visual Detection of Autocorrelation
- 5.7 Test for Normality
- 5.8 Certain Special Cases
 - 5.8.1 Regression through the Origin
 - 5.8.2 Change in Origin and Scale of the Variables
 - 5.8.3 Linearisation of Certain Non-Linear Forms
 - 5.8.4 Choice of Functional Form
- 5.9 Limitations of Regression Analysis
- 5.10 Let Us Sum Up
- 5.11 Key Words
- 5.12 Some Useful Books/References
- 5.13 Answers/Hints to Check Your Progress Exercises

5.0 OBJECTIVES

After going through this unit, you should be in a position to

- appreciate the importance of model diagnostics in econometrics and its role in ensuring the reliability of regression analysis results;
- determine techniques for analysing residuals such as plotting residuals against predicted values and over time to detect outliers
- assess the estimated regression model in terms of goodness of fit.
- analyse visual methods for detecting heteroscedasticity and autocorrelation in regression models;
- explore testing for normality; and
- address special cases in regression analysis such as regression through the origin, variable origin and scale adjustments, and optimal functional form selection.

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5.1 INTRODUCTION

The pursuit of estimates for parameters that underlie economically meaningful relationships has consistently stood at the core of applied econometric modelling endeavours. Consequently, estimation of parameters is an important step in econometrics.

Within the framework of the Classical Linear Regression Model (CLRM), a pivotal assumption stipulates that the model employed for analysis is “correctly specified” (see Unit 8 for details on specification issues). This requirement presents a formidable challenge, as the notion of a flawless model remains elusive. In the realm of econometric modelling, the objective is to encapsulate the essential characteristics of an economic phenomenon while considering elements such as underlying economic theory, prior empirical research, intuition, and research expertise. Attempting to encompass every conceivable factor that impacts a particular research subject would lead to the creation of an impractical and cumbersome model with restricted practical applicability.

It is not always possible to build a perfect econometric model at the first instance. After estimation of the parameters we check whether the model satisfies the assumptions and whether the model is specified properly. We will discuss specification issues in Unit 8. In this Unit we will try to examine the model on the basis of certain rudimentary methods such as visual inspection of the residuals.

5.2 ISSUES IN ESTIMATION

We have studied the four assumptions regarding the error term in a simple regression model. These assumptions are: $E(u_i) = 0$; $E(u_i^2) = \sigma^2$; $E(X_i u_i) = 0$; $E(u_i u_j) = 0$. These assumptions are ideal scenarios – they are not fulfilled in all the cases. In particular, when $E(u_i^2) \neq \sigma^2$, we encounter the problem of heteroscedasticity (i.e., the error variance is not constant throughout the dataset). When $E(X_i u_i) \neq 0$, we have the problem of endogeneity (i.e., the explanatory variable is not independent of the error term). When $E(u_i u_j) \neq 0$, we face the problem of autocorrelation in the dataset. When the assumptions are violated the OLS estimators do not fulfil the desirable properties of an estimator.

There are test procedures to detect the problems of heteroscedasticity, endogeneity and autocorrelation. However, by looking into the residual plot of a regression model, we can sense the possibility of certain problems in the dataset.

When computing the Ordinary Least Squares (OLS) estimate of the parameter ‘ β ’, certain observations might wield disproportionate influence compared to others. Should a handful of observations exert an exceptionally strong impact, it is prudent to investigate whether they stem from erroneous data, such as decimal point misplacement, or pertain to atypical cases, such as including the CEO of Apple in your sample on monthly salary of individuals! In a broader context, it is worthwhile to assess the robustness of the estimation results concerning seemingly minor alterations in the sample. In specific instances, opting for more

robust estimation techniques may prove advisable over employing OLS. Given that a linear regression model may not always suit the data at hand, it is essential to gauge its suitability by defining residuals and scrutinizing residual plots.

5.3 ANALYSIS OF RESIDUALS

Recall that we add an error variable (u) to the regression model, which is not measurable. Based on the estimated coefficients of the model, we obtain the residuals ($\hat{u}$). The difference between the observed value of the dependent variable (Y) and the predicted value ($\hat{Y}$) is called the residual ($\hat{u}$). Each data point has one residual.

Residual = (Observed value – Predicted value)

In terms of notations

$$\hat{u}_i = (Y_i - \hat{Y}_i) \quad \dots (5.1)$$

As you know from Unit 3, the error variable (u) is assumed to have zero mean. The sum of the residuals also is equal to zero. We denote the residual as $\hat{u}$ or e . That is, $\sum \hat{u}_i = 0$. You can also write it as $\sum e = 0$.

Residuals can be regarded as the disparities between the actual data points and corresponding model predictions. Consequently, these discrepancies serve as a measure of the unaccounted-for variability in the dependent variable, attributable to the limitations of the regression model.

5.3.1 Residual Plot

A high value of the error variance (σ_u^2) indicates lower explanatory power of a regression model (recall the definition of R^2). High value of the error variance can occur if the residuals are widely dispersed. Therefore, we should find out if any of the residuals has a very high value. For this purpose, we should examine the residual plot of a regression model. First, we estimate the simple regression model ($Y_i = \alpha + \beta X_i + u_i$) and then we compute the residuals.

A residual plot is a graphical representation where the horizontal axis represents the independent variable (X_i), and the vertical axis displays the residuals ($\hat{u}_i$). If the data points in a residual plot exhibit a random and evenly spread distribution around the horizontal axis, a linear regression model is a suitable choice for the data. Conversely, suppose the data points in the residual plot do not exhibit such random pattern – then it suggests that a nonlinear model would be a better alternative.

5.3.2 Detection of Potential Problems

Conducting a visual examination of residuals stands out as a highly efficient method for assessing the appropriateness of a regression model's fit and verifying its foundational assumptions. Different types of graphical representations are available for scrutinizing various assumptions, and these visual aids can be generated using econometric software.

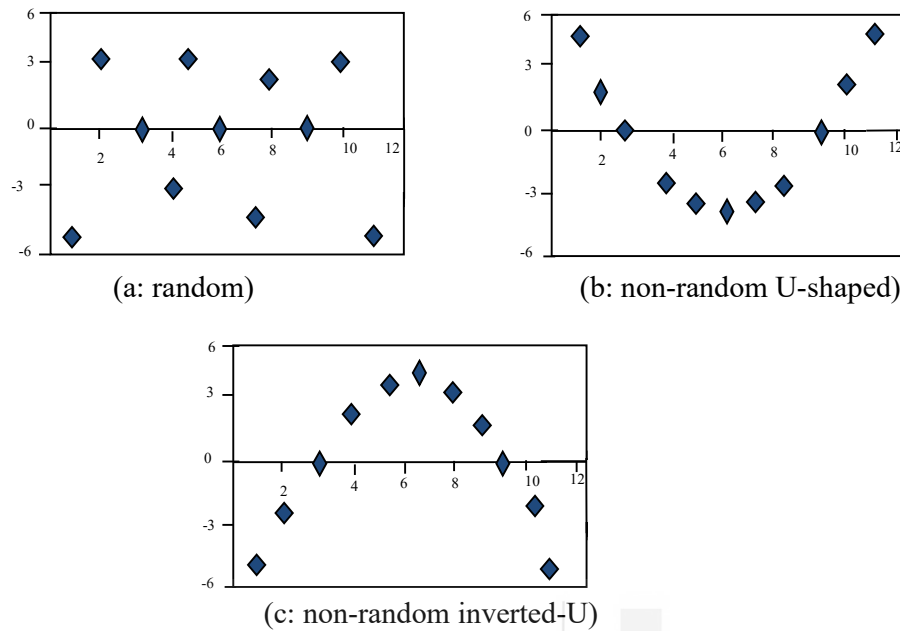


Fig. 5.1: Patterns of Residual Plot

In Fig. 5.1 we present various patterns observed in a residual plot. There are three patterns shown in the figure. The first one (a) indicates a random pattern. The second one (b) indicates a U-shaped pattern while the third one (c) indicates an inverted-U pattern. Thus, the second and the third patterns are non-random. If we notice some non-random pattern in the residual plot, we infer that the linear regression model is not a good fit.

5.3.2 Residuals against Predicted Values

Creating a plot of residuals (e.g., $\hat{u}_i$ or any of the scaled residuals (see subsequent section on scaling the variables) against their corresponding predicted or fitted values ($\hat{Y}_i$) proves valuable in identifying various prevalent model deficiencies. Different interpretations can be attributed to specific types of plots featuring $\hat{Y}_i$ versus $\hat{u}_i$.

If the residual plot exhibits a pattern where the residuals can be enclosed within a horizontal band (see Fig. 5.2(a)), and these residuals fluctuate somewhat randomly within that band, it signifies the absence of model flaws. On the other hand, if the plot displays a pattern where the residuals are enclosed within an outward-opening funnel shape (see Fig. 5.2(b)), then the error variance is not constant. The error variance (σ_u^2) increases with the predicted values ($\hat{Y}_i$). When examining the plots and observing a pattern where the residuals can fit within an inward-opening funnel (see Fig. 5.2(c)), this pattern implies that the error variance is not uniform but rather decreases as a function of the predicted values ($\hat{Y}_i$).

In the event that the plot exhibits a configuration where the residuals can fit within a double bow shape (see Fig. 5.2(d)), this particular pattern serves as an indicator that the error variance is not uniform but instead is related to the predicted values ($\hat{Y}_i$) in a proportional manner, ranging between 0 and 1. In such cases, Y might follow a ‘binomial distribution’ and suggests that the presumed relationship between Y and the predictor variables (X_s) is nonlinear.

When the plot reveals a pattern where the residuals are enclosed within a curved shape (see Fig. 5.2(e)), this signifies nonlinearity in the model. It suggests that

the presumed relationship between the dependent variable (Y) and the independent variables (X_s) is nonlinear. This type of residual plot could also imply the need for incorporating additional explanatory variables into the model. In some cases, applying transformations (such as logarithmic) to the variables could prove beneficial when addressing such nonlinearity.

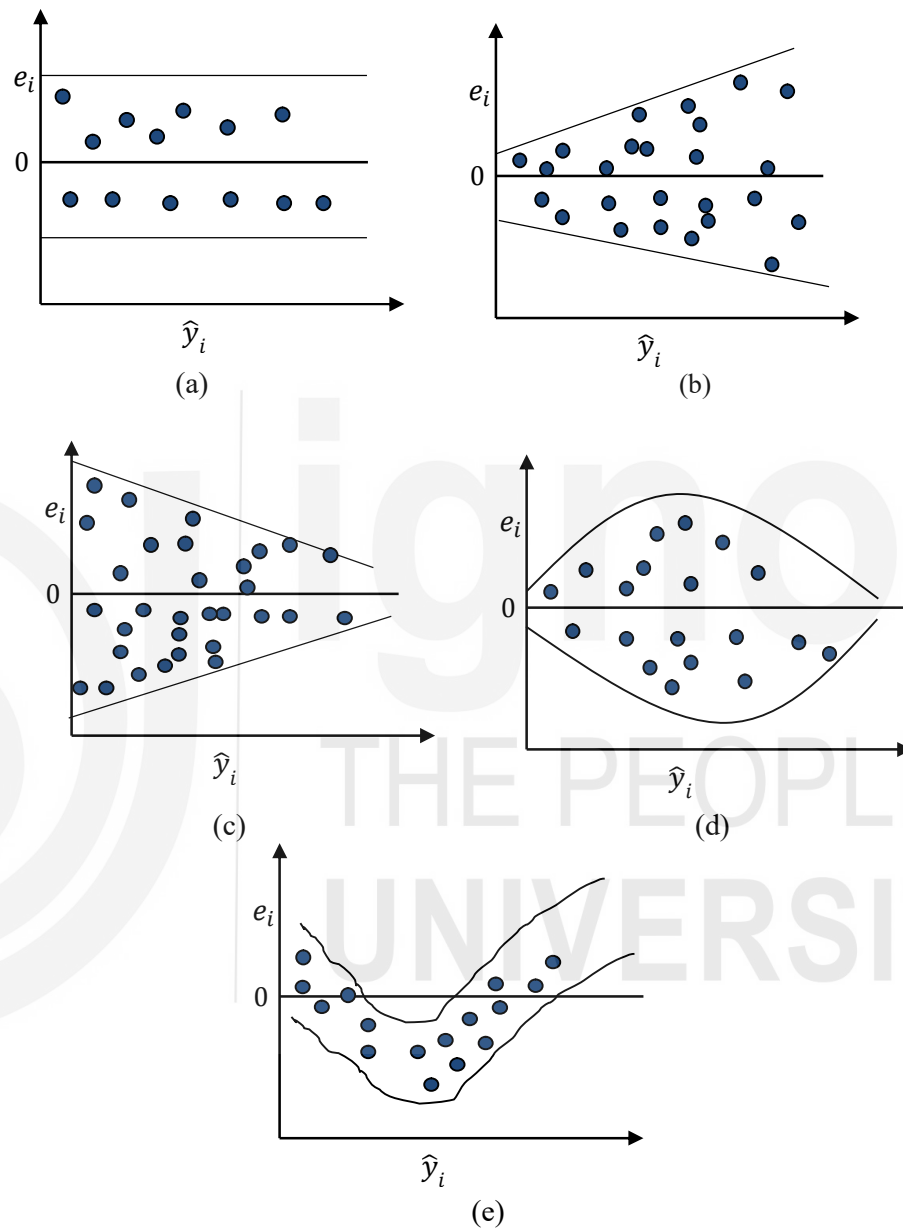


Fig. 5.2: Residual Plot

Plotting residuals against their corresponding explanatory variables can also provide valuable insights. The interpretation of these plots aligns with that of residual-versus- $\hat{Y}_i$ plots and can be summarized as follows:

- (a) If all the residuals cluster within a horizontal band, with fluctuations appearing randomly within this band, it signifies a desirable scenario with no apparent model defects.

- (b) When the residuals take an outward-opening funnel shape or an inward-opening funnel shape, this pattern suggests that variance is not constant across the sample.
- (c) If the plot shows a double bow pattern or nonlinear trend, it implies that the assumed relationship between the dependent variable (Y_i) and the explanatory variable (X_i) may not be accurate. In such cases, various possibilities emerge, including y possibly being a proportion, the need for higher-order terms in X_i , or the requirement for data transformation to be considered.

5.3.4 Residual Plots over Time

Knowing the temporal sequence of data collection allows for the generation of residual plots with respect to time. Taking into account the Y-axis residuals and arranging them with respect to time is analogous to plotting residuals against $\hat{Y}_i$. In lieu of $\hat{Y}_i$, simply use the time order on the X-axis.

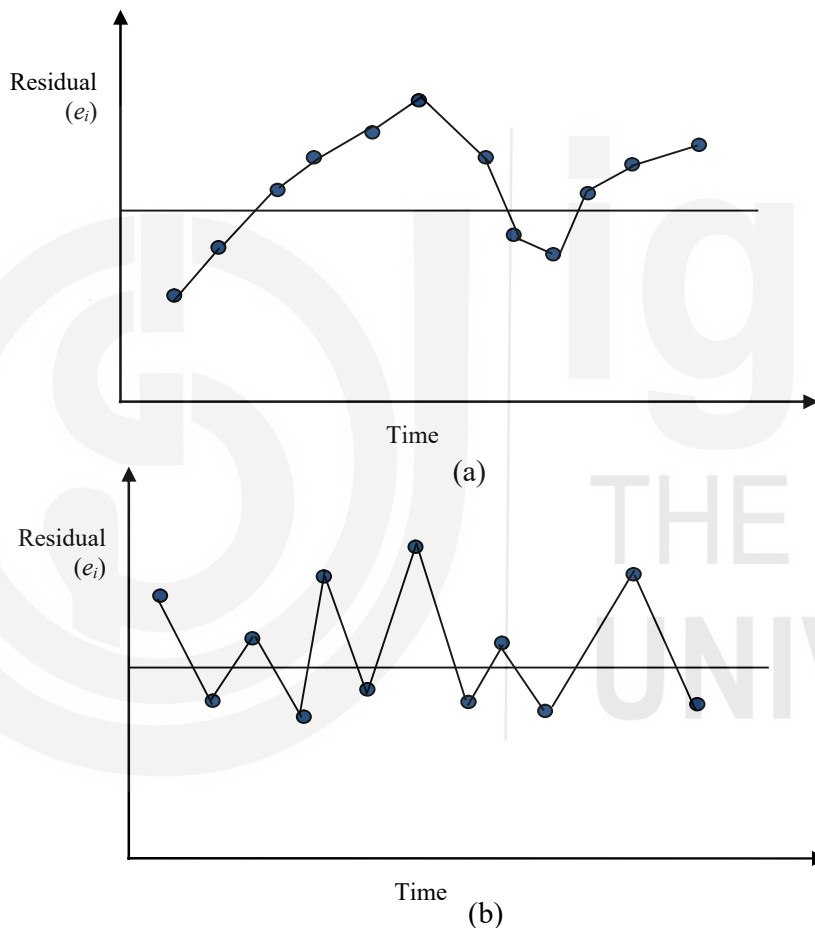


Fig. 5.3: Inter-temporal Residual Plots

If all the residuals are confined within a horizontal band, and they exhibit a somewhat random fluctuation within this band, it is considered favourable and suggests the absence of obvious model defects. A funnel shape that opens outward or inward suggests that the variance is not consistent but rather changing over time. If you observe a double bow pattern or a nonlinear pattern, it implies that the assumed relationship is not linear. In such instances, it is advisable to incorporate linear or quadratic time terms into the model.

The time-ordered residual plot may reveal a connection between errors occurring at one time point and errors at other time points. This connection between model errors across various time periods is termed autocorrelation.

Fig.5.3(a) shows the presence of positive autocorrelation while the presence of negative autocorrelation is demonstrated in Fig. 5.3(b)

5.4 OUTLIERS

In econometrics, outliers are data points that significantly differ from the majority of the observations in a dataset. They can have a substantial impact on statistical analyses and regression model, potentially leading to inaccurate parameter estimates and misleading conclusions. Outliers can occur for various reasons, such as data entry errors, measurement errors, or genuine extreme observations.

Let us try find out the conceptual and graphical explanation of outliers in econometrics: As you know, outliers are observations that fall far away from the typical pattern of data points in a dataset. In econometrics, outliers can manifest in different ways:

- (a) **Positive Outliers:** These are observations that have values much higher than the majority of data points. Positive outliers can be caused by extraordinary events or conditions, such as a sudden spike in sales due to a promotion or a financial crisis leading to extreme stock price changes.
- (b) **Negative Outliers:** Negative outliers are observations with values significantly lower than the typical data points. They might be due to errors or unusual circumstances, such as a company reporting a loss much larger than expected.
- (c) **Multivariate Outliers:** In some cases, outliers might not be apparent when considering a single variable but become noticeable when you examine the relationships between multiple variables. Multivariate outliers can have a substantial impact on regression models.

In regression analysis, an outlier is defined as an observation having a notably large residual (represented as $\hat{u}_i$), which stands out when compared to the residuals of the other observations. In a simple bivariate regression, identifying such a significant residual is straightforward due to its substantial vertical deviation from the estimated regression line. It is important to note that there might be multiple outliers present. Alternatively, one can examine the value of u_i^2 to address the issue of signs (as the residuals can be either positive or negative). Among the various visualization tools available, scatter plot and box plot are the most preferred for outlier detection.

SCATTER PLOT

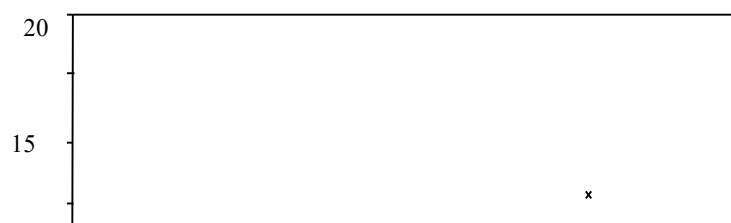


Fig. 5.4: A Scatter Plot showing Outlier at X=375

The scatter plot in Fig 5.4 reveals a basic linear relationship between X and Y for most of the data, and a single outlier (at X = 375).

The concern with outliers lies not only in their divergence from the bulk of the data but rather in their potential to exert a disproportionate influence on estimation techniques such as ordinary least squares. In such instances, an outlier can be categorized as an ‘influential observation’.

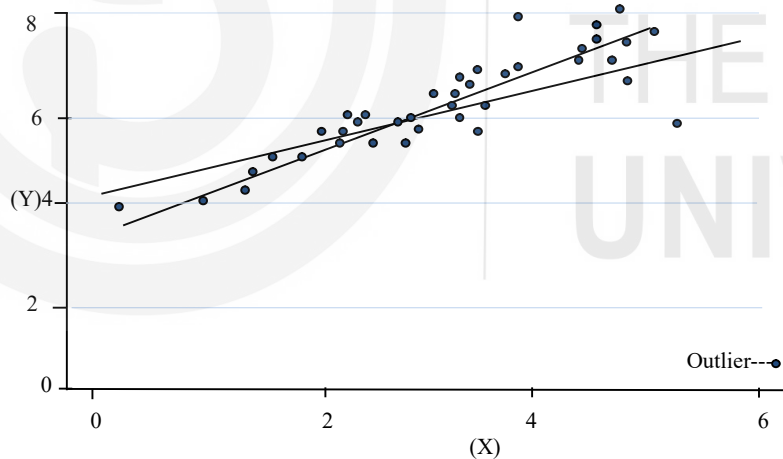


Fig. 5.5: Outlier Effects on Estimated Regression Line

One initial approach to assess the potential existence of outliers in a regression analysis involves examining the ordinary least squares (OLS) residuals, using all available observations. However, it is important to note that this method may not always provide conclusive insights. Remember that OLS relies on minimizing the sum of squared residuals as a fundamental principle, this suggests that when residuals are larger in magnitude, they incur a greater penalty.

Thus, the ordinary least squares (OLS) method aims to mitigate the presence of exceptionally large residuals. This is exemplified by the significant impact an outlier, as depicted in the figure below, can have on the estimated regression line. Consequently, it is advisable to scrutinize an observation's residual when estimating model coefficients using only the remaining portion of the dataset.

Check Your Progress 1

- 1) Why is it important to assess the influence of individual observations on the Ordinary Least Squares (OLS) estimation process, and what steps can be taken to address potential issues?

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- 2) Explain how residual plots can help in detecting potential issues such as heteroscedasticity and autocorrelation in regression models.

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- 3) Why is it important to detect and address outliers in regression analysis? How do outliers affect the techniques such as Ordinary Least Squares (OLS)?

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5.5 VISUAL DETECTION OF HETEROSCEDASTICITY

To assess the presence of heteroscedasticity, conduct a visual examination of residuals plotted against fitted values, or alternatively, plot the independent variable suspected to be associated with the variance of the stochastic error variable. In a large sample, an ideal scenario would reveal a consistent 'envelope' of uniform width when plotting residuals against the independent variable. In contrast, smaller samples may exhibit slightly larger residuals near the mean of the distribution compared to the extremes. Therefore, if the residuals appear to have a consistent size across all values of X , or in the case of a small sample, exhibit slightly larger residuals near the mean of X , it is generally reasonable to assume that heteroscedasticity is not a significant concern. However, if the

residual plot displays an uneven envelope, where the width of the envelope significantly varies for different values of X , it is advisable to proceed with a more formal test for heteroscedasticity.

In the analysis of regression results, it is crucial to verify that the residuals exhibit consistent variance. When the residuals display different values of variance across segments we can sense the presence of heteroscedasticity (see Fig. 5.6(a)).

On the contrary, when the residuals maintain a uniform variance, we refer to this condition as homoscedasticity. Homoscedasticity indicates that the residuals possess equal variability across all independent variables [see Fig. 5.6(b)].

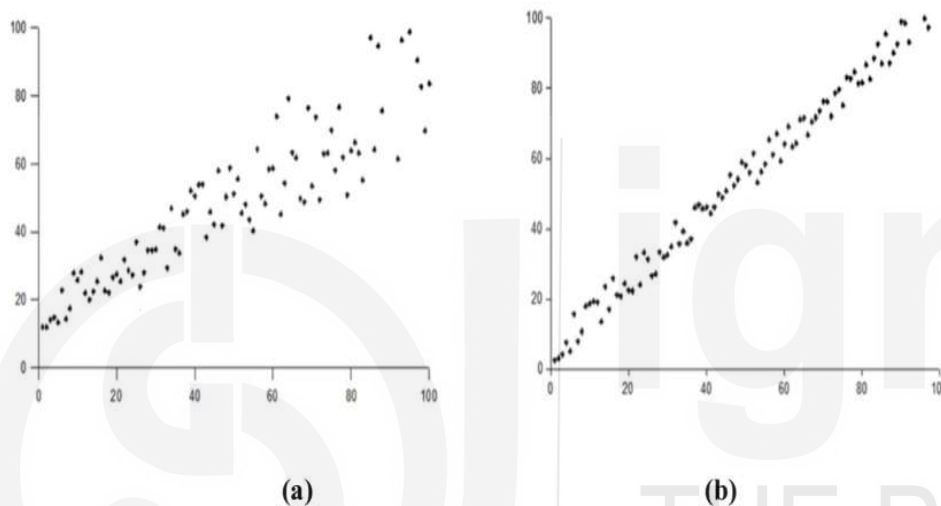


Fig. 5.6: Heteroscedasticity

In the case of a homoscedastic model, we can reasonably assume that the residuals originate from a population with stable variance. This condition aligns with one of the fundamental assumptions of ordinary least squares (OLS) regression, contributing to a more accurate model.

5.6 VISUAL DETECTION OF AUTOCORRELATION

Autocorrelation, also known as serial correlation, occurs when the errors or residuals in a time series or regression model are correlated with their past values (lagged values). Detecting autocorrelation visually involves examining patterns in the residuals over time:

- (a) **Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) Plots:** These plots are commonly used in time series analysis. ACF measures the correlation between the residuals at different lags, while PACF accounts for the correlation at each lag while removing the influence of shorter lags. Significant spikes or patterns in these plots can suggest autocorrelation.
- (b) **Lag Plots:** Create a scatterplot of the residuals against their lagged values (residuals at previous time points in time series data). If you see points clustering together or forming a pattern, it may indicate autocorrelation.
- (c) **Time Series Plots:** For time series data, simply plotting the residuals over time and visually inspecting for any trends or repeating patterns can provide insights

into autocorrelation. If the Time Sequence Plot displays no discernible pattern, it suggests the absence of autocorrelation (as shown in Fig. 5.7(e)). However, if the plot reveals a pattern, it indicates the presence of autocorrelation, as illustrated in Fig. 5.7.

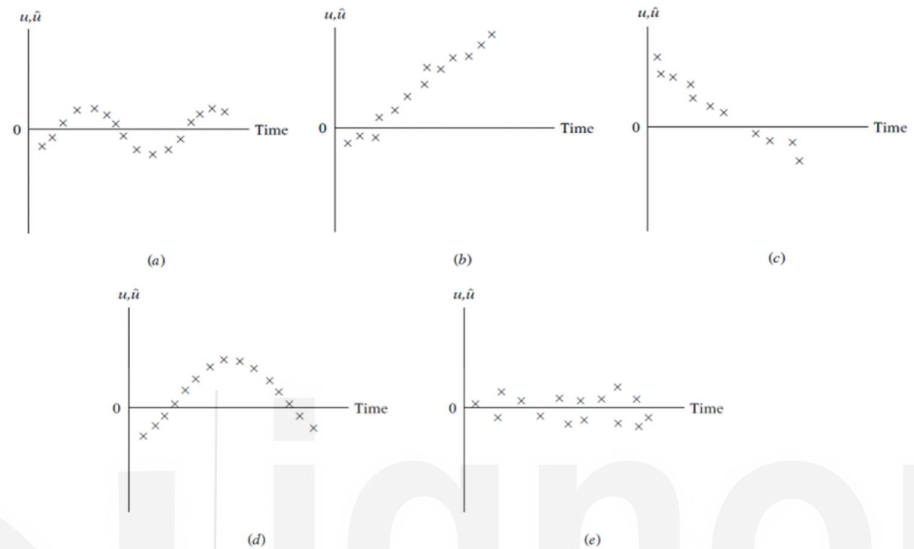


Fig. 5.7: Presence of Autocorrelation

Visual detection of heteroscedasticity and autocorrelation is a preliminary step in model diagnostics. If you suspect the presence of these issues based on visual examination, it is important to follow up with formal statistical tests to confirm and quantify the extent of these problems in your data or model residuals.

5.7 TEST FOR NORMALITY

In the classical normal linear regression model (CNLRM), which is an extension of the Classical Linear Regression Model (CLRM), one of the critical assumptions is that the error term, represented as u_i in the regression model, follows a normal distribution. This assumption holds particular significance when dealing with relatively small sample sizes, as widely used significance tests such as t-test and F-test rely on the normality assumption. Therefore, it becomes essential to assess whether the error term conforms to a normal distribution.

The easiest way to check whether the errors are normally distributed is to use the P-P (Probability-Probability) plot. A P-P plot assesses how closely a theoretical distribution models a given data distribution. A normal probability plot of the residuals is a scatter plot with the theoretical percentiles of the normal distribution on the x-axis and the sample percentiles of the residuals on the y-axis. The comparison line is the 45 degree diagonal line. The diagonal line which passes through the lower and upper percentiles of the theoretical distribution provides a visual aid to help assess whether the relationship between the theoretical and sample percentiles is linear. The distributions are equal if and

only if the plot falls on this line. Any deviation from this line indicates a difference between the two distributions.

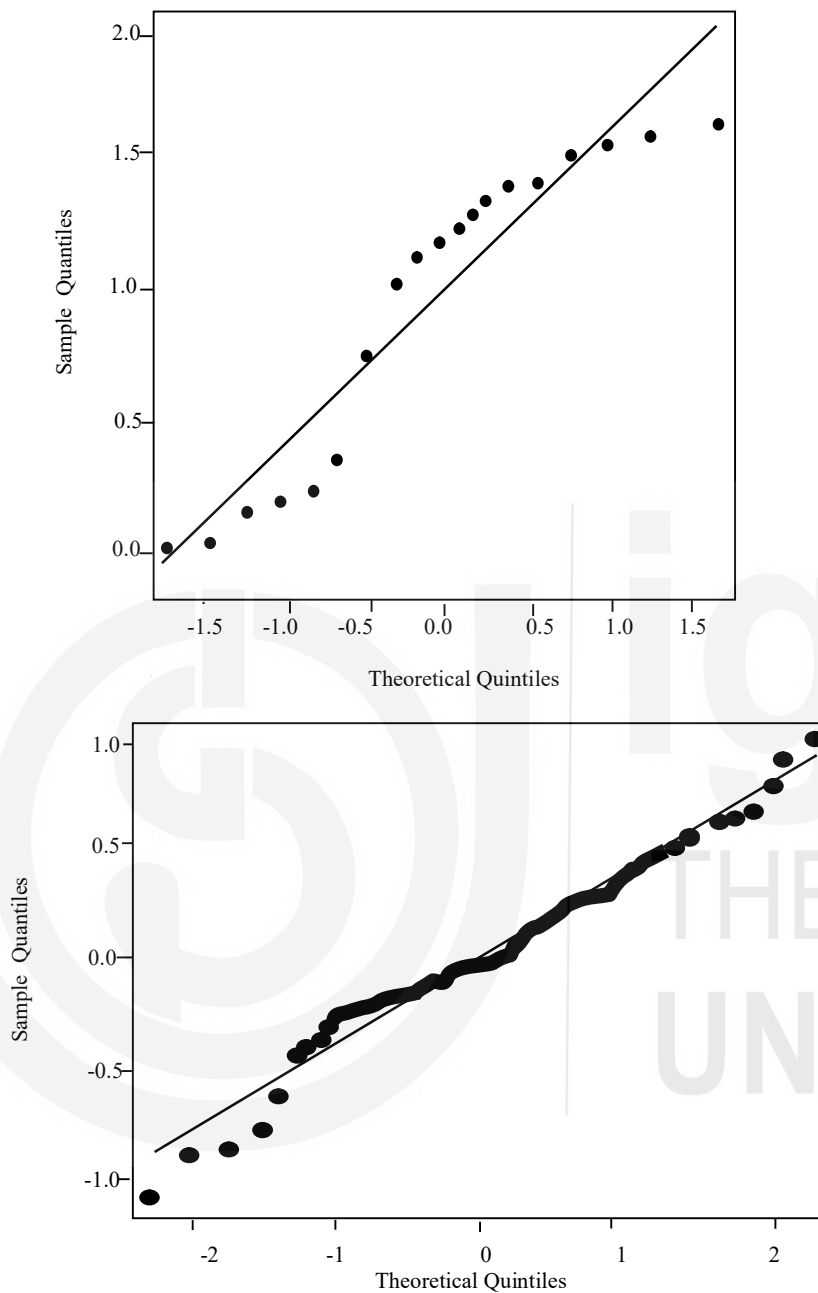


Fig. 5.8: Assessing Normality Using Probability Plot

Fig. 5.8(a) is an example of the P-P plot where the error distribution is not normal while Fig. 5.8(b) is the P-P plot where the error distribution is approximately normal.

Various formal tests of normality are available, with the Jarque–Bera (JB) test being one of the most commonly employed. However, it is important to note that the JB test is designed for large samples and may not be suitable for smaller sample sizes.

The test statistic JB is calculated using the sample skewness and kurtosis. The formula for the Jarque-Bera test statistic is as follows:

$$JB = \frac{n}{6} \left(\frac{S^2}{6} + \frac{(K-3)^2}{24} \right) \quad \dots (5.2)$$

where:

n is the sample size.

S is the sample skewness.

K is the sample kurtosis.

In equation (5.2), if the data follows a normal distribution, JB should be approximately equal to zero. Departures from zero indicate a departure from normality. JB follows a chi-squared distribution with *two* degrees of freedom ($df = 2$) under the null hypothesis of normality. Therefore, you can compare the calculated JB statistic to the chi-squared distribution to determine whether the data significantly deviates from normality.

Typically, if JB statistic exceeds a critical value from the chi-squared distribution at a specified significance level (e.g., 0.05 or 0.01), it suggests that the data significantly departs from a normal distribution, indicating non-normality.

5.8 CERTAIN SPECIAL CASES

In this section we discuss certain issues on regression models, as you can see from the sub-section headings.

5.8.1 Regression through the Origin

While the majority of regression models typically incorporate a constant or intercept term, occasionally, economic theory or logical reasoning may imply that the intercept term (b_0) should equal zero. In such cases, it is worth discussing Ordinary Least Squares (OLS) estimation when the intercept is constrained to be zero. In practical terms, this means that you are modelling the relationship between variables without allowing for a constant offset in the form of an intercept. For instance, the intercept-free linear model

$$Y_i = \beta X_i + u_i \quad \dots (5.3)$$

is equivalent to the model

$$Y_i = \alpha + \beta X_i + u_i$$

when the intercept ' α ' is taken to be zero.

An important characteristic of the zero-intercept model is that it employs raw sums of squares and cross-product terms. In this context, "raw" means that these terms are not adjusted by subtracting the mean values of the variables. This stands in stark contrast to the OLS regression model with an intercept, where the sums of squares and cross-product terms are mean-adjusted, meaning they are represented as deviations from the mean values of the variables.

Unless strong theoretical justifications exist, it is generally not advisable to omit the intercept term in a regression model, for several compelling reasons. Firstly, in a regression model with an intercept term, the sum of residuals always equals zero, whereas this is not guaranteed in a zero-intercept regression model. Secondly, the R-squared values for models with and without intercepts are not directly comparable. In the model with an intercept, R-squared represents the proportion of variation explained by the regression relative to the total variation around the mean of Y. In contrast, for the no-intercept model, R-squared signifies the proportion of variation explained relative to the variation around the origin (i.e., zero).

5.8.2 Change in Origin and Scale of the Variables

Typically, it is straightforward to discern how the intercept and slope estimates change when the units of measurement for the dependent variable are altered. If the dependent variable undergoes multiplication by a constant 'c' – meaning each sample value is multiplied by 'c' – then the OLS intercept and slope estimates likewise multiply by 'c'. This assumption holds when no alterations occur with the independent variable. In general, altering the units of measurement for only the independent variable does not have any bearing on the intercept. As discussed in the previous section, we defined R-squared as a goodness-of-fit metric for OLS regression. Also note that the model's goodness-of-fit remains unaffected regardless of changes in variable units.

Regression coefficients are independent of change of origin but not of scale. By origin, we mean that there will be no effect on the regression coefficients if any constant is subtracted from the value of X and Y. By scale, we mean that if the value of X and Y is either multiplied or divided by some constant, then the regression coefficients will also change.

The sensitivity of regression results to the unit of measurement varies; certain outcomes are affected, while others remain unaffected. Certain important points to note include- First, R-squared value remains constant across these regressions as R-squared is a dimensionless measure, independent of the units in which both the dependent variable (Y) and the independent variable (X) are gauged. Second, the intercept term consistently aligns with the units of measurement of the dependent variable. It is crucial to recall that the intercept signifies the dependent variable's value when the independent variable is zero. Third, when both Y and X are measured using identical units, the slope coefficients, along with their standard errors, stay consistent. However, there are variations in the intercept values and their corresponding standard errors. Fourth, in instances where the Y and X variables are expressed in distinct units, the slope coefficients exhibit dissimilarity, yet the interpretation remains unaltered.

Therefore, the units of measurement for the dependent variable (Y) and the explanatory variables (the X's) influence the interpretation of the regression coefficients. To circumvent this issue, we can represent all variables in a standardized form, where standardization is achieved by subtracting the mean value of each variable from its individual values and then dividing the difference

by the standard deviation of that variable. A noteworthy characteristic of a standardized variable is that its mean is consistently zero, and its standard deviation always equals 1.

5.9 LIMITATIONS OF REGRESSION ANALYSIS

Even when we possess theoretically sound variables that can explain a phenomenon and we manage to gather data on these variables, it is a common challenge not to have a clear understanding of the functional relationship between the dependent variable and the independent variables (see data generation process in Unit 3). In two-variable models, the scatter diagram frequently allows for an assessment of the functional relationship. However, in multiple regression models, identifying the suitable functional form becomes challenging, as visually interpreting scatter diagrams in multiple dimensions is not feasible. Consequently, the stochastic disturbances u_i play a crucial role in regression analysis, a significance we will delve into as we advance to subsequent units. Undoubtedly, selecting a suitable model for empirical estimation demands a considerable amount of skill and experience. However, certain guidelines can be provided:

The literature and theoretical paradigms must be re-visited as the foundational theory might propose a specific type of functional form. It is advisable to determine both the rate of change (slope) of the dependent variable concerning the independent variable and the elasticity of the dependent variable concerning the independent variable.

The parameters selected for the model should align with specific *a priori* expectations. For instance, when examining the demand for laptops in relation to price and other variables, it is reasonable to anticipate a negative coefficient for the price variable.

Occasionally, a given set of data may be reasonably well-fitted by more than one model. R squared values may help in deciding which model is better in some such cases. However, there are some caveats to be considered – as we will explore in the upcoming sections, the inclusion of more regressors in the model leads to an increase in R^2 . What holds greater significance is the theoretical foundation of the chosen model, the signs of the estimated coefficients, and its statistical significance? If the performance of a model is better on the basis of these criteria, a lower value of R^2 does not matter much.

When considering the limitations of regression analysis, several important factors should be taken into account:

Linearity Assumption: It is crucial to assess whether the relationship between the variables under investigation truly follows a linear pattern. This assumption must be carefully examined because deviations from linearity can lead to inaccurate conclusions.

Sensitivity to Outliers: Regression analysis can be highly sensitive to outliers, which are data points that deviate significantly from the overall trend. Researchers must exercise caution when interpreting results, as outliers can

disproportionately influence regression coefficients and may not accurately represent the underlying relationship.

Preliminary Data Visualization: Prior to conducting regression analysis, it is advisable to visualize the data through plots and graphs. This step helps identify potential outliers and provides insights into the overall data distribution. Decisions regarding outlier handling and data transformations should be made based on this preliminary exploration.

Extrapolation Risks: Extrapolation or making predictions outside the range of the explanatory variable, should be approached with caution. Extending regression results beyond the observed data range can lead to unreliable predictions and is often not appropriate. For example, using regression to predict outcomes for a population significantly different from the study sample may yield inaccurate results.

Hidden Variables and Confounding Factors: Researchers should be aware of the presence of lurking variables, also known as confounding or hidden variables. These unaccounted-for factors can distort the relationship between the variables included in the regression model. When confounding variables are present, the observed relationship may be misleading or spurious. Careful consideration and control of potential confounders are essential in regression analysis.

In summary, regression analysis offers valuable insights into relationships between variables, but it is not without limitations. Researchers must critically assess the linearity assumption, address the impact of outliers, visualize data before analysis, exercise caution with extrapolation, and remain vigilant for hidden variables to ensure the reliability and validity of their regression results.

Check Your Progress 2

- 1) Explain how you can visually detect the presence of heteroscedasticity in regression analysis. What are the characteristics of a residual plot indicating heteroscedasticity.

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- 2) Write down the visual methods for detecting autocorrelation in time series or regression models. How do autocorrelation function plots and lag plots help in identifying autocorrelation?

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- 3) Under what circumstances would you consider omitting the intercept term in a regression model?

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- 4) Discuss the implications of a zero-intercept model and the differences in interpretation compared to models with intercepts.

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5.10 LET US SUM UP

Ensuring accurate model specification is crucial in econometric analysis, as it allows for a focused exploration of relevant factors impacting the research theme. Challenges inevitably arise in the quest for a perfect econometric model, underscoring the need for accurate parameter estimation and thorough validation checks, to uphold the underlying assumptions.

Residual analysis emerges as a pivotal tool in assessing model adequacy, with various patterns in residual plots offering valuable insights into model fit. By scrutinizing these patterns, issues such as heteroscedasticity and autocorrelation can be identified and subsequently addressed in an effective manner. Furthermore, the impact of outliers and variable units on regression coefficients is highlighted in the unit. Conducting tests to detect outliers, assess normality, and handle specific scenarios in regression analysis, such as regression through the origin, adjusting variable origins and scales, and selecting the most suitable functional form, significantly contributes to accurate model diagnostics. Finally, certain potential challenges in regression analysis are discussed. By navigating these challenges and leveraging diagnostic techniques, economists can enhance the robustness and reliability of their regression analyses.

5.11 KEY WORDS

- Residual** : It refers to the difference between the observed value and the predicted value in regression analysis.
- Homoscedasticity** : It refers to the presence of uniform variance of residuals across all values of the independent variable.

- Model Diagnostics : Techniques for assessing the adequacy and assumptions of regression models.
- Normality : It refers to the assumption that the error term in a regression model follows a normal distribution.
- Endogeneity Problem : It occurs when an explanatory variable in a regression model is correlated with the error term, violating the assumption of exogeneity.
- Functional Form : The mathematical expression that defines the relationship between the dependent and independent variables in an economic model.
- Economic Model : It refers to the representation of relationship between a response variable (dependent variable) and the predictor variable(s) (independent variable(s)), typically specified by subject or area experts to capture relevant economic phenomena.

5.12 SOME USEFUL BOOKS/REFERENCE

Gujarati, D. (2014). *Econometrics by Examples*. Bloomsbury Publishing.

Verbeek, M. (2017). *A guide to modern econometrics*. John Wiley & Sons.

Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data*. MIT Press.

Links:

Berman H.B., "Residual Analysis in Regression". Available at: <https://stattrek.com/regression/residual-analysis> [Accessed on Date: 9/6/2023].

Econometric Data Analysis: A Cost-Effective, Largely Successful Methodology By Lee S. Friedman. Available at: <https://gsppi.berkeley.edu/lee-friedman/documents/econometric-data-analysis-2.pdf>

IIT, Kanpur: <http://home.iitk.ac.in/~shalab/regression/Chapter4-Regression-ModelAdequacyChecking.pdf>

5.13 ANSWERS/ HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) You should mention the impact of influential observations on estimation results and provide examples of situations where such observations might occur. Also, touch upon the concept of robust estimation techniques and their relevance in mitigating these issues (Refer to Section 5.1 & Section 5.2)

**Classical Regression
Model**

- 2) Refer to the patterns in residual plots and how they indicate the presence of specific problems.
- 3) Think about the influence outliers can have on the estimation process and the potential biases they introduce into regression results.

Check Your Progress 2

- 1) Consider the patterns in residual plots and how they differ between homoscedastic and heteroscedastic models.
- 2) Refer to the graphical representations and patterns in residuals that indicate autocorrelation given in Section 5.6
- 3) Reflect on how a zero-intercept model alters the interpretation of coefficients and affects the model's fit and predictions compared to models with intercepts.
- 4) Refer to Sub-Section 5.8.1 and answer.



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UNIT 6 ESTIMATION OF MULTIPLE REGRESSION MODELS*

Structure

- 6.0 Objectives
- 6.1 Introduction
- 6.2 Specification of the Model
 - 6.2.1 Matrix Representation
 - 6.2.2 Classical Assumptions
- 6.3 OLS Method of Estimation
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 - 6.2.3 Estimator of Error Variance
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 - 6.4.1 Algebraic Properties
 - 6.4.2 Statistical Properties
- 6.5 Best Linear Unbiased Estimator (BLUE)
- 6.6 Let Us Sum Up
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- 6.9 Answers/Hints to Check Your Progress Exercises

6.0 OBJECTIVES

After going through this Unit you should be in a position to

- explain the concept of multiple regression models with two or more explanatory variables;
- represent multiple regression model in matrix form;
- apply ordinary least squares (OLS) method to find out the estimators;
- describe the properties of the OLS estimators; and
- prove that OLS estimators are the best linear unbiased estimator (BLUE).

6.1 INTRODUCTION

In Unit 4 we discussed the relationship between two variables (one explained, one explanatory) using the simple regression model. We considered the dependent

variable Y to be a function of the independent variable X . However, in practise we notice that there is more than one explanatory variable in a regression model. In such situations it is difficult to draw conclusion on how X affects Y *ceteris paribus* as it is unrealistic to assume that all other factors influencing Y are not correlated to X .

Multiple regression models provide an amicable solution to such a problem. As the name suggests, we consider two or more explanatory variables (and one dependent variable) in multiple regression models. Examining the effects of a particular independent variable on the dependent variable *ceteris paribus* becomes possible as with multiple regression models we can control for several other factors simultaneously which could have influence on the dependent variable. This enables us to test economic theories as well as evaluate policy impacts. Several explanatory variables can be accommodated in multiple regression models allowing us to draw inferences on causality. Multiple regression models enable us to add suitable factors to the model resulting in explanation of more variation in the dependent variable. Such benefits of a multiple regression model make it one of the most widely used empirical tools for analysis in economics and social sciences. In the following section we will develop an understanding about a multiple regression model and evaluate the advantages of using it for analysis.

6.2 SPECIFICATION OF THE MODEL

In multiple regression models we have one explained variable and many explanatory variables (say, k). The multiple regression model is specified as follows:

$$Y_i = \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \dots + \beta_k X_{ki} + u_i \quad (i = 1, 2, \dots, n) \quad \dots (6.1)$$

Here, the subscript ' i ' denotes the observation number in the sample, while the subscript ' k ' denotes the explanatory variable. Thus, Y_i is the i^{th} observation of the dependent variable, $\beta_1, \beta_2, \dots, \beta_k$ are the parameters, $X_{1i}, X_{2i} \dots X_{ki}$ are the i^{th} observations on the explanatory variables, and u_i is the stochastic error term. Specification of the multiple regression model is similar to that of the two-variables model, except that there is more than one explanatory variable here. Generally, we assume the first explanatory variable to be unity ($X_1 = 1$) for all the observations. Under such an assumption, the multiple regression model becomes

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \dots + \beta_k X_{ki} + u_i \quad (i = 1, 2, \dots, n) \quad \dots (6.2)$$

The first parameter, β_1 , in (6.2) is the intercept that we discussed in Unit 4. To avoid confusion in notation, very often we drop the superscript ' i ', and represent multiple regression model in the following form:

$$Y = \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + u \quad \dots (6.3)$$

As in the case of two-variable regression model, the multiple regression model is estimated by the ordinary least squares (OLS), method of moments, and the maximum likelihood method. We will discuss the OLS and maximum likelihood methods of estimation in this Unit. In both the methods, the approach is the same as mentioned in Unit 4. In the OLS method, for example, we must minimise the

sum of squared residuals ($\sum \hat{u}_i^2$) and take partial derivatives with respect to the parameters. Thus, there are k normal equations (see Unit 4) for a regression model with k independent variables. Finding equilibrium estimates of all the parameters in the model is a difficult and time-consuming process. In order to simplify the process of finding equilibrium values, we take the help of matrix algebra.

6.2.1 Matrix Representation

In Unit 3 of this course, we provided an overview of matrix algebra. Let us express equation (6.2) as follows:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{U} \quad \dots (6.4)$$

where

$$\begin{array}{cccc} \mathbf{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} & \mathbf{X} = \begin{bmatrix} X_{11} & X_{12} & \dots & X_{1k} \\ \vdots & \vdots & \dots & \vdots \\ X_{n1} & X_{n2} & \dots & X_{nk} \end{bmatrix} & \boldsymbol{\beta} = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} & \mathbf{U} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \\ (n \times 1) & (n \times k) & (k \times 1) & (n \times 1) \end{array}$$

The dependent variable $\mathbf{Y}$ is a $(n \times 1)$ vector. The explanatory variables are denoted by matrix $\mathbf{X}$ which has the order $(n \times k)$. As you can see, i^{th} row indicates the i^{th} observation and j^{th} column indicates j^{th} explanatory variable. Thus, X_{ij} represents the value of the j^{th} explanatory variable of the i^{th} observation. Vector $\boldsymbol{\beta}$ denotes the set of explanatory variables. As there are k explanatory variables in the model, the order of $\boldsymbol{\beta}$ vector is $(k \times 1)$. The vector $\mathbf{U}$ denotes the stochastic error variable, which has an order of $(n \times 1)$. These notations are commonly used in econometrics. You should keep track of the dimensions of the matrices.

6.2.1 Classical Assumptions

Recall the four assumptions regarding the error variable (u). These are: (i) $E(u_i) = 0$, (ii) $E(u_i^2) = \sigma^2$, (iii) $E(u_i u_j) = 0$ for $i \neq j$, and (iv) $E(X_i u_i) = 0$. We need to have corresponding assumptions for the multiple regression model. The errors are identically and independently distributed with mean zero and variance σ^2 . The explanatory variables are non-stochastic and independent of the error terms. The explanatory variables are linearly independent. Thus,

These assumptions are

$$(i) \quad E(\mathbf{U}) = \mathbf{0} \quad \dots (6.5)$$

Assumption (6.5) says that $E(u_i) = 0$; $i = 1, 2, \dots, n$. The expected value of u_i is zero.

$$(ii) \quad E(\mathbf{U}\mathbf{U}') = \sigma^2 \mathbf{I} \quad \dots (6.6)$$

Assumption (6.6) is a precise way of expressing two assumptions: $E(u_i u_j) = 0$ and $E(u_i^2) = \sigma^2$. Let us define the variance-covariance matrix $\mathbf{U}\mathbf{U}'$. Note that it has a dimension of $(n \times n)$. It is given by

$$\begin{bmatrix} \text{var}(u_1) & \text{cov}(u_1 u_2) & \dots & \text{cov}(u_1 u_n) \\ \vdots & \vdots & \dots & \vdots \\ \text{cov}(u_n u_1) & \text{cov}(u_n u_2) & \dots & \text{var}(u_n) \end{bmatrix} = \begin{bmatrix} \sigma^2 & 0 & \dots & 0 \\ \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \dots & \sigma^2 \end{bmatrix}$$

In the above matrix, the diagonal elements are σ^2 and the off-diagonal elements are zero.

(iii) X has full rank ... (6.7)

It implies that there is no exact relationship among the explanatory variables. Thus, none of the explanatory variables is a linear combination of other explanatory variables. If there is an exact relationship between any two explanatory variables, there will be a decline in the rank of matrix X . The rank of matrix X is and $k < n$. An implication of assumption (6.7) is that the matrix $X'X$ is invertible, that is, $(X'X)^{-1}$ exists.

(iv) $E(X'U) = 0$... (6.8)

Assumption (6.8) implies that none of the explanatory variables is related with the error variable u . In other words, the error variable U cannot be predicted on the basis of the independent variables. In order to satisfy this assumption, we assume that X is non-stochastic (i.e., fixed in repeated samples). In Unit 13 we will relax this assumption and assume that some of the regressors are stochastic in nature.

By assumption $\text{var}(u_1) = \text{var}(u_2) = \dots = \text{var}(u_n) = \sigma^2$. Thus, variance of the error term is constant across all the observations. This feature of the error term is called homoscedasticity. When the error variance does not remain constant across observations (that is, $\text{var}(u_1) \neq \text{var}(u_2) \neq \dots \neq \text{var}(u_n)$) we have a situation of heteroscedasticity, which we will discuss in Unit 11. Similarly, we assumed that $E(u_i u_j) = 0$. It implies that the error term of one observation is not correlated with the error term of another observation. In other words, there is no autocorrelation.

6.3 OLS METHOD OF ESTIMATION

In the two-variable case we minimised the residual sum of squares (RSS), given by $\sum \hat{u}_i^2$. In the multiple regression model, we have to minimise the RSS for obtaining the estimators. We mentioned above in equation (6.4) that the regression model is given by

$$Y = X\beta + U$$

The estimated regression model is given by

$$\hat{Y} = X\hat{\beta} \quad \dots (6.9)$$

Therefore,

$$\hat{U} = Y - X\hat{\beta} \quad \dots (6.10)$$

The RSS is given by $\hat{U}'\hat{U}$ which has a dimension of $(1 \times n)(n \times 1) = (1 \times 1)$. Thus, RSS is a scalar quantity.

From equation (6.10) we have

$$\hat{U}'\hat{U} = (Y - X\hat{\beta})'(Y - X\hat{\beta}) \quad \dots (6.11)$$

On simplification, (6.11) can be written as

$$\hat{U}'\hat{U} = Y'Y - \hat{\beta}'X'Y - Y'X\hat{\beta} + \hat{\beta}'X'X\hat{\beta}$$

Since $\hat{\beta}'X'Y$ and $Y'X\hat{\beta}$ are scalar quantities (the order of both the matrices is (1×1)), we can write

$$\hat{U}'\hat{U} = Y'Y - 2\hat{\beta}'X'Y + \hat{\beta}'X'X\hat{\beta} \quad \dots (6.12)$$

In the least squares method we choose such value of $\hat{\beta}$ that minimises the **RSS**. The conditions of finding a minimum are: (i) first derivative should be equal to zero, and (ii) second derivative should be positive.

Let us take the derivative of $\hat{U}'\hat{U}$ with respect to $\hat{\beta}$.

$$\frac{\partial \hat{U}'\hat{U}}{\partial \hat{\beta}} = 0 - 2X'Y + 2X'X\hat{\beta} \quad \dots (6.13)$$

While taking derivative of (6.12) with respect to $\hat{\beta}$, you should pay special attention to derivative of the term $\hat{\beta}'X'X\hat{\beta}$. Both $\hat{\beta}'$ and $\hat{\beta}$ are present here. We have taken its derivative is $2X'X\hat{\beta}$, and not $2\hat{\beta}'X'X$. Let us look into the following example.

Example 6.1

Suppose we have two vectors: **a** and **x**.

$$a = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Thus,

$$a'x = a_1x_1 + a_2x_2 + a_3x_3$$

Let us take derivative of $a'x$ with respect to x .

$$\frac{\partial a'x}{\partial x} = \frac{\partial a'x}{\partial \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}} = \frac{\partial}{\partial \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}} a_1x_1 + a_2x_2 + a_3x_3 = \begin{bmatrix} a_1 + 0 + 0 \\ 0 + a_2 + 0 \\ 0 + 0 + a_3 \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = a$$

Coming back to equation (6.13), we eliminate $\hat{\beta}'$ (and not $\hat{\beta}$) while taking the derivative of $\hat{U}'\hat{U}$ with respect to $\hat{\beta}$. By equating $\frac{\partial \hat{U}'\hat{U}}{\partial \hat{\beta}} = 0$, we obtain

$$-2X'Y + 2X'X\hat{\beta} = 0$$

Or,

$$X'X\hat{\beta} = X'Y \quad \dots (6.14)$$

Equation (6.14) represents the 'normal equations' of the least squares method.

If we pre-multiply equation (6.14) by $(X'X)^{-1}$, we obtain

$$\hat{\beta} = (X'X)^{-1}X'Y \quad \dots (6.15)$$

Let us take the second derivative of $\hat{U}'\hat{U}$ with respect to $\hat{\beta}$. It is similar to taking the derivative of (6.13). Thus,

$$\frac{\partial^2 \hat{U}'\hat{U}}{\partial \hat{\beta} \partial \hat{\beta}'} = 2 X'X \quad \dots (6.16)$$

Since $X'X$ is a positive definite matrix, we find that $\hat{\beta}$ represents the minimum of the RSS.

The estimated regression model is given by

$$\hat{Y} = X\hat{\beta} \quad \dots (6.17)$$

6.3.1 Mean of OLS Estimators

From equation (6.15) we know that

$\hat{\beta} = (X'X)^{-1} X'Y$. By substituting the value of Y from equation (6.1), we obtain

$$\hat{\beta} = (X'X)^{-1} X'(X\beta + U) = (X'X)^{-1} X'X\beta + (X'X)^{-1} X'U$$

Thus,

$$\hat{\beta} = \beta + (X'X)^{-1}X'U \quad \dots (6.18)$$

$$\text{Mean of } \hat{\beta} = E(\hat{\beta}) = E[\beta + (X'X)^{-1}X'U]$$

Since X is kept fixed in repeated samples, it is not stochastic or random. Further, by assumption $E(U) = 0$

Therefore,

$$E(\hat{\beta}) = \beta + (X'X)^{-1}X'E(U) = \beta \quad \dots (6.19)$$

The expected value of the OLS estimator $\hat{\beta}$ is the true parameter β . An implication of the above is that the OLS estimators are *unbiased*.

6.3.2 Variance of OLS Estimators

Now let us find out the variance of the OLS estimator $\hat{\beta}$. Recall that $\hat{\beta}$ is a $k \times 1$ vector of the estimates. The values of the estimates, vary across samples. The variance of the OLS estimators $\hat{\beta}$ is given by

$$\text{var}(\hat{\beta}) = E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'] \quad \dots (6.20)$$

In fact, (6.19) represents the variance-covariance matrix of the parameters in the regression model. It is a matrix of the order $(k \times k)$. The diagonal terms of this matrix represent the variance of the parameters while the off-diagonal terms represent the covariance between the parameters (for example, the ij^{th} term of the matrix $[(\hat{\beta} - \beta)(\hat{\beta} - \beta)']$ represents the covariance between β_i and β_j).

$$\text{From (6.18), we find that } \hat{\beta} - \beta = (X'X)^{-1}X'U$$

Therefore,

$$E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'] = E[(X'X)^{-1}X'UU'X(X'X)^{-1}]$$

$$= (X'X)^{-1}X'E(UU')X(X'X)^{-1}$$

Since $E(UU') = \sigma^2 I$ by assumption, we have

$$\text{var}(\hat{\beta}) = \sigma^2(X'X)^{-1} \quad \dots (6.21)$$

Note that σ^2 is a constant and the order of the symmetric matrix $(X'X)$ is k .

6.3.3 Estimator of Error Variance

For finding the variance of $\hat{\beta}$ we need an estimator of σ^2 . As you know, the error variable cannot be observed. We take the residuals as a proxy of the error variable. Let us find out the variance of the stochastic error in terms of the residuals.

In Unit 4, while dealing with two-variable regression models, we found that an unbiased estimator of the error variance is given by $RSS/(n - 2)$. Recall that there are two parameters (intercept and slope) in the two-variable regression model. We will generalise the above result to cases where the number of explanatory variables is k .

We know that the residual is given by $\hat{U} = (Y - \hat{Y})$. Let us substitute the value of $\hat{Y}$ from equation (6.17).

$$\begin{aligned} \hat{U} &= Y - X\hat{\beta} = Y - X(X'X)^{-1}X'Y \\ &= [I - X(X'X)^{-1}X']Y \\ &= MY \end{aligned} \quad \dots (6.22)$$

where $M = [I - X(X'X)^{-1}X']$

Remember that M is an 'idempotent matrix'. Thus, M is a symmetric matrix of order n . Further, if we multiply M with itself, its value remains unchanged. Thus,

$$M = M'$$

and

$$\begin{aligned} MM &= [I - X(X'X)^{-1}X'] [I - X(X'X)^{-1}X'] \\ &= [I - X(X'X)^{-1}X' - X(X'X)^{-1}X' + X(X'X)^{-1}X'X(X'X)^{-1}X'] \\ &= I - X(X'X)^{-1}X' - X(X'X)^{-1}X' + X(X'X)^{-1}X' \\ &= I - X(X'X)^{-1}X' = M \end{aligned}$$

Further,

$$MX = [I - X(X'X)^{-1}X']X = X - X(X'X)^{-1}X'X = X - X = 0 \quad \dots (6.23)$$

From (6.22) we have $\hat{U} = MY$. We can write MY as

$$MY = M(X\beta + U) = MX\beta + MU = 0 + MU = MU \quad \dots (6.24)$$

The RSS is given by $\sum \hat{u}_i^2$ which can be written in matrix form as $\hat{U}'\hat{U}$. You should note the difference between the matrix $U'U$ and the matrix UU' . The matrix UU' (of the order $(n \times n)$) represents the variance-covariance of the error term. On the other hand, matrix $U'U$ (of the order (1×1)) is a scalar quantity.

$$\hat{U}'\hat{U} = (MU)'(MU) = U'M'MU = U'MU \quad \dots (6.25)$$

Note that $\mathbf{U}'\mathbf{M}\mathbf{U}$ is a scalar as the order of the matrix is $(1 \times n)(n \times n)(n \times 1) = (1 \times 1)$. Further, recall the concept of trace of a matrix discussed in Unit 2. Trace of a matrix is the sum of the diagonal elements of the matrix. Thus, trace of a scalar is the scalar itself. Let us point out two properties of the trace of a square matrix.

- (a) For matrix $\mathbf{A}$, we have $Tr(\mathbf{A}) = \sum a_{ii}$
- (b) For two matrices $\mathbf{A}$ and $\mathbf{B}$, we have $Tr(\mathbf{AB}) = Tr(\mathbf{BA})$

By taking expected value of (6.23), we obtain

$$E(\hat{\mathbf{U}}'\hat{\mathbf{U}}) = E(\mathbf{U}'\mathbf{M}\mathbf{U}) = E[Tr(\mathbf{U}'\mathbf{M}\mathbf{U})] = E[Tr(\mathbf{U}'\mathbf{M}\mathbf{U})]$$

$$E(\mathbf{U}'\mathbf{M}\mathbf{U}) = E[Tr(\mathbf{U}'\mathbf{M}\mathbf{U})] = E[Tr(\mathbf{U}\mathbf{U}'\mathbf{M})]$$

(Assume that $\mathbf{U}'\mathbf{M} = \mathbf{A}$ and $\mathbf{U} = \mathbf{B}$)

Thus,

$$\begin{aligned} E[Tr(\mathbf{U}\mathbf{U}'\mathbf{M})] &= \sigma^2 Tr(\mathbf{M}) \\ &= \sigma^2 Tr[\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'] = \sigma^2 Tr(\mathbf{I}) - \sigma^2 Tr[\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'] \\ &= \sigma^2 Tr(\mathbf{I}) - \sigma^2 Tr[\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}] \\ &= \sigma^2 n - \sigma^2 k \\ &= \sigma^2 (n - k) \end{aligned}$$

Thus,

$$E(\hat{\mathbf{U}}'\hat{\mathbf{U}}) = \sigma^2 (n - k)$$

By re-arranging terms, we obtain an estimator of σ^2 . Let us call it $\hat{\sigma}^2$.

$$\hat{\sigma}^2 = E(\hat{\mathbf{U}}'\hat{\mathbf{U}})/(n - k) = \mathbf{RSS}/(n - k) \quad \dots (6.26)$$

Check Your Progress 1

- 1) Bring out the advantages of multiple regression models over bivariate regression models.

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- 2) Derive the expression for the OLS estimator in the following linear regression model:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{U}.$$

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- 3) Derive the expression for the variance of the OLS estimator in the following regression model; $Y = X\beta + U$.

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- 4) Why is it important that the X matrix should have full rank in the regression model; $Y = X\beta + U$?

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6.4 PROPERTIES OF OLS ESTIMATORS

A good estimator should satisfy certain statistical properties such as unbiasedness, consistency and efficiency. In this section we will examine whether the OLS estimator fulfils these properties. In addition, there are certain algebraic properties of the OLS estimators. We begin with the algebraic properties.

6.4.1 Algebraic Properties

- (i) The normal equations of the regression model are given by

$$(X'X)\hat{\beta} - X'Y = 0$$

Let us substitute the value of Y from equation (6.1) in the above equation, we get

$$(X'X)\hat{\beta} - X'(X\hat{\beta} + \hat{U}) = 0$$

$$\text{Or, } (X'X)\hat{\beta} - (X'X)\hat{\beta} + X'\hat{U} = 0$$

Or,

$$X'\hat{U} = 0 \quad \dots (6.27)$$

Equation (6.25) implies that the residual and the explanatory variables are not correlated.

- (ii) The predicted value $\hat{Y}$ and the residual $\hat{U}$ are not correlated.

$$\hat{Y}'\hat{U} = (X\hat{\beta})'\hat{U} = \hat{\beta}'X'\hat{U} = 0 \quad \dots (6.28)$$

6.4.2 Statistical Properties

Let us discuss about the statistical or theoretical properties of OLS estimators.

- (a) **Unbiasedness:** In the regression equation, we have added an error variable (U), which is random. The addition of a random variable to the regression model makes the dependent variable (Y) also random. Since the estimator ($\hat{\beta}$) is a linear combination of the dependent variable, the estimator will provide estimates which are random. Unbiasedness property of an estimator can be checked by taking repeated samples. It says that the average of the estimates will be equal to the parameters if we take repeated samples. In technical terms,

the expected value of the estimator is equal to the parameter, i.e., $E(\hat{\beta}) = \beta$. This we have proved in equation (6.18). Thus, the OLS estimators are unbiased.

- (b) **Consistency:** The consistency property of an estimator is different from unbiasedness. An estimator is said to be consistent if its value approaches the parameter value as the sample size is increased. We know from equation (6.17) that $\hat{\beta} = \beta + (X'X)^{-1}X'U$. To prove that $\hat{\beta}$ is consistent, we have to prove that the second term on the right-hand side has a probability limit (*plim*) of zero. This term is the product of two terms $(X'X)^{-1}$ and $X'U$. Neither $(X'X)^{-1}$ nor $X'U$ has a probability limit. However, $(n^{-1} \times n) = 1$. We can write $(X'X)^{-1}X'U$ as $\left(\frac{1}{n}X'X\right)^{-1} \frac{1}{n}X'U$. Therefore,

$$plim(\hat{\beta}) = \beta + \left(plim \frac{1}{n}X'X\right)^{-1} plim \frac{1}{n}X'U = \beta \quad \dots (6.29)$$

- (c) **Efficiency:** efficiency of an estimator refers to the ‘minimum variance unbiased estimator’. Since the variance of the estimator is the lowest, the estimates will be scattered closer to the parameter. Consequently, the estimator will give better and more accurate estimates than other estimators having higher variance. Notice, the comparison is among unbiased estimators only. As we will see below, the OLS estimator is the ‘best linear unbiased estimator’ (BLUE).

6.5 BEST LINEAR UNBIASED ESTIMATOR (BLUE)

We are given a sample data set, which contains data on X and Y . We maintain the classical assumptions about the stochastic error term U . We have two estimators: the OLS estimator $\hat{\beta}$, and any other linear unbiased estimator $\tilde{\beta}$. We have to prove that the OLS estimator $\hat{\beta}$ is the best.

This property of the OLS estimator is also called the ‘Gauss-Markov theorem’. The OLS estimator is given by

$$\hat{\beta} = (X'X)^{-1}X'Y \quad \dots (6.14)$$

Here, the X matrix is assumed to be kept fixed in repeated samples. Further, we know the value of X from the sample of data available to us. We can write $\hat{\beta} = (X'X)^{-1}X'Y = WY$ where $W = (X'X)^{-1}X'$. Thus, $\hat{\beta}$ is a linear combination of Y and $\hat{\beta}$ is a linear estimator.

In order to prove that $\hat{\beta}$ is BLUE, we compare it with another *linear unbiased* estimator, say, $\tilde{\beta}$. Since $\tilde{\beta}$ is a linear estimator, we can consider

$$\tilde{\beta} = CY \quad \dots (6.30)$$

By substituting the value of Y from the regression model (6.1), we can write (6.30) as

$$\tilde{\beta} = C(X\beta + U) = CX\beta + CU \quad \dots (6.31)$$

Note that we are comparing amongst unbiased estimators. Thus, we should have

$$E(\tilde{\beta}) = \beta \quad \dots (6.32)$$

By using the value of $\tilde{\beta}$ from equation (6.30), we obtain

$$E(\tilde{\beta}) = E(CX\beta + CU) = CX\beta + CE(U) \quad \dots (6.33)$$

From the above, $E(\tilde{\beta}) = \beta$ (i.e., $\tilde{\beta}$ is unbiased) only if we assume that

$$CX = I \quad \dots (6.34)$$

Let us find out the variance of $\tilde{\beta}$. We can re-write equation (6.31) as

$$\tilde{\beta} - \beta = CU \quad \dots (6.35)$$

The variance of $\tilde{\beta}$ is given by

$$var(\tilde{\beta}) = E[(\tilde{\beta} - \beta)(\tilde{\beta} - \beta)'] \quad \dots (6.36)$$

By using (6.34), we can re-write (6.35) as

$$var(\tilde{\beta}) = E[(CUU'C)] = C E(UU')C' = \sigma^2 CC' \quad \dots (6.37)$$

The variance of the OLS estimator $var(\hat{\beta}) = \sigma^2(X'X)^{-1}$ whereas the variance of any other linear unbiased estimator $var(\tilde{\beta}) = \sigma^2 CC'$. Since $\hat{\beta}$ and $\tilde{\beta}$ are linear estimators, let

$$C = (X'X)^{-1}X' + D \quad \dots (6.38)$$

where D is a $(k \times n)$ matrix (i.e., the same order as the $(X'X)^{-1}X'$ matrix).

Let us multiply equation (6.37) by X .

$$CX = [(X'X)^{-1}X' + D]X = (X'X)^{-1}X'X + DX = I + DX$$

From equation (6.33), we know that $CX = I$. Therefore,

$$DX = 0 \quad \dots (6.39)$$

Remember that $var(\hat{\beta}) = \sigma^2(X'X)^{-1}$ and $var(\tilde{\beta}) = \sigma^2 CC'$. Let us compare between CC' and $(X'X)^{-1}$.

$$\begin{aligned} CC' &= [(X'X)^{-1}X' + D][(X'X)^{-1}X' + D]' \\ &= [(X'X)^{-1}X' + D][X(X'X)^{-1} + D'] \\ &= (X'X)^{-1}(X'X)(X'X)^{-1} + DX(X'X)^{-1} + (X'X)^{-1}X'D' + DD' \end{aligned}$$

Since $DX = X'D' = 0$, we obtain

$$CC' = (X'X)^{-1} + DD' \quad \dots (6.40)$$

Since DD' is positive semi-definite,

$$CC' \geq (X'X)^{-1} + DD' \quad \dots (6.41)$$

Thus, $var(\hat{\beta})$ is less than $var(\tilde{\beta})$. In other words, the OLS estimators are the best linear unbiased estimators (BLUE).

Check Your Progress 2

- 1) Prove that the residual and the dependent variable are not correlated in the case of the multiple regression models.

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- 2) Prove that the OLS estimators are the best linear unbiased estimators (BLUE).

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- 3) Show that the OLS estimators are unbiased, consistent and efficient.

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6.6 LET US SUM UP

Multiple regression models enable us to include several explanatory variables in the regression model. Further, it allows specification of a general functional form on the relationship between dependent variable and independent variables. Because of its benefits, the multiple regression models are widely used.

In this Unit we specified the multiple regression model in its matrix form. We applied the method of ordinary least squares (OLS) and derived the estimators of the model. The desirable properties of the estimators such as unbiasedness, consistency and efficiency are discussed. The OLS estimators $\hat{\beta}$ are the best linear unbiased estimator (BLUE) of β .

6.7 KEY WORDS

Asymptotic Properties	: It refers to the properties of estimators and test statistics that apply when the sample size grows without bound.
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- Biased Estimator** : An estimator whose expectation, or sample mean, is different from the population value it is supposed to be estimating.
- Ceteris Paribus*** : It refers to the assumption that all other relevant factors are held fixed.
- Maximum Likelihood Estimation (MLE)** : A broadly applicable estimation method where the estimates are chosen to maximize the log-likelihood function.
- Quadratic Functions** : It refers to the functions that contain squares of one or more explanatory variables; they capture diminishing or increasing effects on the dependent variable.

6.8 SOME USEFUL BOOKS

- Greene, W. H., 2018, “*Econometric Analysis*.” Eighth Edition, Pearson Education
- Johnston, J., Dinardo, J., 2007, “*Econometric Methods*”, 4th Edition, McGraw-Hill Education (Asia), International Edition, pp. 109-125
- Maddala, G. S. and K. Lahiri, 2012, *Introduction to Econometrics*, Fourth Edition, Wiley
- Verbeek, M. 2012, “*A Guide to Modern Econometrics*”, Fourth Edition, John Wiley & Sons

6.9 ANSWERS/ HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) The advantages of using multiple regression model are as follows:
 - a. It enables us to control for several other factors simultaneously which could have influence on the dependent variable.
 - b. It helps us examine the effects of an independent variable on the dependent variable *ceteris paribus*.
 - c. It eliminates the chances of bias arising due to correlation between the error term and the explanatory variable which is so peculiar to two-variable models.
 - d. It presents a general functional form on the cause and effect relationships between variables.
- 2) Refer to Section 6.3 and answer.
- 3) The variance of the OLS estimators is given by $var(\hat{\beta}) = \sigma^2 (X'X)^{-1}$. Please see Sub-Section 6.3.2 for derivation.

- 4) The OLS estimator is given by $\hat{\beta} = (X'X)^{-1}X'Y$. Full rank of the X matrix implies that the explanatory variables are independent of one another. If any of the columns in the matrix is a linear combination of other columns, we cannot find the matrix $(X'X)^{-1}$.

Check Your Progress 2

- 1) Go through Sub-Section 6.4.1 and answer.
- 2) Go through Section 6.5 and answer.
- 3) You have to show that the OLS estimators are unbiased, they are consistent, and they are the best linear unbiased estimator. Go through Section 6.4 and Section 6.5.



UNIT 7 EVALUATION OF MULTIPLE REGRESSION MODELS*

Structure

- 7.0 Objectives
- 7.1 Introduction
- 7.2 Coefficient of Determination
- 7.3 Hypothesis Testing
 - 7.3.1 *t*-Test
 - 7.3.2 Confidence Interval
 - 7.3.3 *F*-Test
- 7.4 Testing Linear Restrictions
 - 7.4.1 Wald Test
 - 7.4.2 Lagrange-Multiplier (LM) Test
 - 7.4.3 Likelihood Ratio (LR) Test
 - 7.4.4 Empirical Exercises
- 7.5 Let Us Sum Up
- 7.6 Key Words
- 7.7 Some Useful Books
- 7.8 Answers/ Hints to Check Your Progress Exercises

7.0 OBJECTIVES

After going through this Unit, you will be able to

- explain the concept of ‘Coefficient of Determination’ with an example;
- outline the procedure for testing the statistical significance of a single parameter in an estimated regression model;
- describe how the statistical significance of a group of parameters can be assessed by using the *F*-test; and
- Discuss the tests for linear restrictions, viz., Wald, Lagrange Multiplier, and Likelihood Ratio tests.

7.1 INTRODUCTION

In the previous Unit, we discussed the specification of multiple regression models. We also derived the estimators of the parameters by applying the OLS and maximum likelihood methods of estimation. In this unit, we will first consider the importance of the concept of Coefficient of Determination given by R^2 . We then proceed to learn how to test for the significance of the estimated parameters for individual regressors. We will also learn how to interpret the results. The extent to which the independent variables together explain the variation in the dependent variable will also be seen. We finally learn about three procedures for testing the linear restrictions.

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7.2 COEFFICIENT OF DETERMINATION

In Unit 4, while presenting the simple regression model, we derived the coefficient of determination, denoted by R -square (R^2). Recall that R^2 is defined as the ratio of ‘Explained Sum of Squares’ (ESS) to ‘Total Sum of Squares’ (TSS). In multiple regression models, we can define the coefficient of determination in the same manner.

Regression, as you know, is an analytical tool to explain the variation in the dependent variable (Y) in terms of the changes in the independent variables (X_i ’s). We can decompose the difference between Y_i and its arithmetic mean $\bar{Y}$ as follows:

$$(Y_i - \bar{Y}) = (Y_i - \hat{Y}_i) + (\hat{Y}_i - \bar{Y}) \quad \dots (7.1)$$

Let us look into the intuitive idea behind the R^2 . The expected value of variable Y is $\bar{Y}$. Therefore, in the absence of the regression model, the expected value of Y is $\bar{Y}$. When we try to explain Y through a regression model, the expected value of Y is $\hat{Y}$. Therefore, the difference between $\hat{Y}_i$ and $\bar{Y}$, $(\hat{Y}_i - \bar{Y})$, is due to the explanatory variables included in the regression model and can be called as the ‘explained variation’. The difference $(Y_i - \hat{Y}_i)$ denotes the part that cannot be explained by the regression model. Hence, squaring both sides of equation (7.1) and summing over all observations (1 to n), we obtain

$$\sum (Y_i - \bar{Y})^2 = \sum (Y_i - \hat{Y}_i)^2 + \sum (\hat{Y}_i - \bar{Y})^2 \quad \dots (7.2)$$

Total Variation	Residual Variation	Explained Variation
Or, TSS = (Total Sum of Squares)	RSS + (Residual Sum of Squares)	ESS (Explained Sum of Squares)

R^2 is defined as

$$R^2 = \frac{ESS}{TSS} = \frac{\sum (\hat{Y}_i - \bar{Y})^2}{\sum (Y_i - \bar{Y})^2} \quad \dots (7.3)$$

Note that R^2 measures the proportion of the variation in Y which is explained by the multiple regression equation. Therefore, R^2 is often used as a goodness-of-fit statistic to compare the validity of regression results under alternative specifications of the independent variables in the model.

MULTIPLE REGRESSION MODELS: TESTING

The value of R -square (R^2) ranges between 0 and 1. An implication of this bound is that R^2 is always positive. Let us consider two hypothetical situations: First, when $ESS = TSS$ (i.e., $RSS = 0$) when the explanatory variables explain the variation in the dependent variable fully. Then, $R^2 = 1$. Second, when $ESS = 0$ (i.e., the explanatory variables do not explain any variation in Y). Then, $TSS =$

RSS and $R^2 = 0$. Thus, the range within which the coefficient of determination remains is $0 \leq R^2 \leq 1$.

The coefficient of determination (R^2) reflects the explanatory power of a regression model. A higher value of R^2 implies higher explanatory power of the underlying regression model compared to some other model. Thus, R^2 can be used as a measure of goodness of fit in multiple regression models. Remember that we can compare two values of R^2 only when the dependent variable is the same. *We cannot compare two regression models if the dependent variable is different.*

On re-arrangement of terms in equation 7.3, we obtain

$$R^2 = \frac{ESS}{TSS} = \frac{TSS - RSS}{TSS} = (1 - \frac{RSS}{TSS})$$

Thus,

$$R^2 = 1 - \frac{\sum \hat{u}_i^2}{\sum (Y_i - \bar{Y})^2} \quad \dots (7.4)$$

From (7.4) we find that as RSS decreases, R^2 increases.

7.2.1 Over-Fitting

Suppose we have a regression model with two explanatory variables, and we obtain R^2 value as 0.63. It implies that the two explanatory variables explain 63 per cent variation in the dependent variable. Now, suppose, we add one more explanatory variable to the regression model, i.e., the number of explanatory variables is three. What will happen to the R^2 value? We can compare the R^2 values of both the regression models since the dependent variable is the same.

It is observed that when we add explanatory variables to a regression equation, the R^2 value increases (i.e., the explanatory power of the model increases). The R^2 value never decreases when we add an explanatory variable. Thus, there is a tendency on the part of researchers to include more explanatory variables in a regression model. In such cases, even if logically not correct and statistically not significant, many explanatory variables get added to the regression model. We cannot trust the inferences drawn from such models! Therefore, a question arises is: where do we stop?

7.2.2 Adjusted- R^2

The degrees of freedom of a regression model with sample size n and number of explanatory variables k (including the intercept) for the three sums of squares are as follows:

$$\text{TSS: } (n - 1) \qquad \text{ESS: } 1 \qquad \text{RSS: } (n - k)$$

There is thus a decrease in the degrees of freedom of RSS when we add explanatory variables to a model. Notice that R^2 as a measure of goodness of fit does not account for the degrees of freedom. A natural solution is to use *variances* (i.e., sums of squares divided by degrees of freedom), and not mere *variations*. It eliminates the dependence of goodness of fit on the number of

independent variables in the model. Such an R^2 is defined as $\bar{R}^2$, and is called as the corrected/adjusted R^2 . In notations, we have

$$\bar{R}^2 = 1 - \frac{RSS/(n-k)}{TSS/(n-1)} \quad \dots (7.5)$$

On re-arrangement of terms in equation (7.5), we obtain

$$\bar{R}^2 = 1 - \frac{RSS (n-1)}{TSS (n-k)} = 1 - \frac{(TSS - ESS) (n-1)}{TSS (n-k)} = 1 - \left[\left(1 - \frac{ESS}{TSS} \right) \frac{(n-1)}{(n-k)} \right]$$

Or,

$$\bar{R}^2 = 1 - (1 - R^2) \frac{(n-1)}{(n-k)} \quad \dots (7.6)$$

where n is number of observations in the regression model and k is number of parameters in the equation. As n becomes large, the difference between $\bar{R}^2$ and R^2 tend to be small.

$\bar{R}^2$ has several properties. One, when new variables are added to a regression model, the R^2 always increases, but the $\bar{R}^2$ may rise or fall. Second, the use of $\bar{R}^2$ eliminates the incentives for researchers to include more variables in a model without logic. Third, $\bar{R}^2$ is always less than or equal to R^2 . Fourth, the $\bar{R}^2$ can also be negative while R^2 is always positive. For these reasons, $\bar{R}^2$ is regarded as a more desirable measure of goodness of fit compared to R^2 . Let us now consider an example on multiple regressions.

Example 7.1:

Consider the regression model $Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + u$. The following data are given. Calculate the values of R^2 and $\bar{R}^2$.

Y	40	60	50	70	90
X_1	4	6	7	10	13
X_2	8	12	10	5	7

We prepare a table as below to find out the arithmetic mean of Y , X_1 and X_2 . Then, we subtract the mean from the respective variables to obtain y , x_1 and x_2 .

Y	X_1	X_2	y	x_1	x_2
40	4	8	- 22	- 4	- 0.4
60	6	12	- 2	- 2	3.6
50	7	10	- 12	- 1	1.6
70	10	5	8	2	- 3.4
90	13	7	28	5	- 1.4

From the above table we obtain the following:

$$\begin{array}{lll} \sum Y = 310 & \sum X_1 = 40 & \sum X_2 = 42 \\ \sum x_1 y = 260 & \sum x_2 y = -84 & \sum x_1 x_2 = -21 \\ \sum x_1^2 = 50 & \sum x_2^2 = 29.70 & \sum y^2 = 1480 \\ n = 5 & & \end{array}$$

Therefore,

$$\begin{aligned} \hat{\beta}_1 &= 5.72 \\ \hat{\beta}_2 &= 1.24 \\ \hat{\alpha} &= 5.86 \end{aligned}$$

Further,

$$RSS = \sum \hat{u}^2 = 96.84$$

The standard error of the estimators are

$$\begin{aligned} s_{\hat{\alpha}} &= 6.96 \\ s_{\hat{\beta}_1} &= 1.18 \\ s_{\hat{\beta}_2} &= 1.54 \end{aligned}$$

Thus, the estimated equation is

$$\hat{Y} = 5.86 + 5.72 X_1 + 1.24 X_2$$

From this equation we can compute

$$\begin{aligned} ESS &= \sum \hat{y}^2 = 1383.16 \\ TSS &= \sum y^2 = 1480.00 \\ R^2 &= \frac{ESS}{TSS} = 0.93 \end{aligned}$$

Adjusted- R^2 is

$$\bar{R}^2 = 1 - (1 - R^2) \frac{(n-1)}{(n-k)}$$

Or,

$$\bar{R}^2 = 1 - (1 - 0.93) \frac{(5-1)}{(5-3)} = 1 - 0.07 \times \frac{4}{2} = 0.86.$$

Check Your Progress 1

- 1) Define the Coefficient of Variation. Indicate its expression for multiple regression models.

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- 2) Why is R^2 often used as a measure of ‘goodness of fit’?
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- 3) Specify an alternative expression for R^2 indicating its relationship with the Residual Sum of Squares. What is the relationship that is conveyed by this expression?
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- 4) Is it always beneficial to add another ‘explanatory variable’ in a multiple regression model? What is a way out of this situation?
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- 5) Explain why the adjusted- R^2 is preferred to R^2 .
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7.3 HYPOTHESIS TESTING

In the previous Unit, we learnt that under the Gauss-Markov assumptions, the OLS estimator b has normal distribution with a mean 0 and covariance matrix $\sigma^2(X'X)^{-1}$. We can use this information to test the hypotheses associated with the unknown population parameter vector β . For testing the significance of b_k we will use the test statistic:

$$z = \frac{b_k - \beta_k}{\sigma \sqrt{c_{kk}}} \quad \dots (7.7)$$

where z is the test statistic;

β_k is the parameter to be tested;

b_k is the estimate of β_k obtained from the sample data; and

c_{kk} is the standard error of b_k .

The variance-covariance matrix of the estimators is given by $\text{var}(\mathbf{b}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$. To obtain the standard error of the estimators, we first compute the inverse of the matrix $(\mathbf{X}'\mathbf{X})$. Second, we multiply it by σ^2 . Third, we take the square root of the matrix $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$. Suppose, the ij^{th} element of this matrix is denoted by σc_{ij} . The diagonal elements provide us the standard error of the individual parameters. For example, the first diagonal element (let us say, σc_{11}) provides us with the standard error of b_1 . Similarly, the second diagonal element (σc_{22}) provides us with $se(b_2)$. Thus, when we test for the statistical significance of b_k , we should take its standard error $\sigma \sqrt{c_{kk}}$ as denominator.

The test statistic z follows standard normal distribution. If we replace σ with its estimate s , it can be shown that the unbiased estimator s^2 is independent of b and follows chi-squared distribution with $(n - k)$ degrees of freedom. The corresponding test statistic is given as

$$\chi^2_{(n-k)} \sim \frac{(n-k)s^2}{\sigma^2} \quad \dots (7.8)$$

We can, therefore, define the random variable t_k to test for b_k as

$$t_k = \frac{b_k - \beta_k}{s \sqrt{c_{kk}}} \quad \dots (7.9)$$

The value of the t -ratio in equation (7.9) is the ratio of a standard normal variable $(b_k - \beta_k)$, and the square root of an independent chi-squared variable $(s \sqrt{c_{kk}})$. It therefore follows the Student's t -distribution with $(n - k)$ degrees of freedom. The t -distribution (see Appendix - B at the end of the book) resembles the standard normal distribution for large values of $(n - k)$. For very large values of $(n - k)$, the two (t -distribution and z -distribution) are identical. The expression (7.9) can be used to further develop 'test statistic' and 'confidence interval'.

7.3.1 t -Test

Testing of a hypothesis involves two steps: (i) construction of a test statistic, and (ii) a decision rule. Let me illustrate it this with an example. Suppose I obtained the estimate b_k . I want to test the hypothesis $H_0: \beta_k = 0$. The alternative hypothesis is $H_1: \beta_k \neq 0$. It implies that I have a two-tailed test. From (7.9) I have the test statistic

$$t_k = \frac{b_k - 0}{se(b_k)} = \frac{b_k}{se(b_k)} \quad \dots (7.10)$$

In (7.10), there are no unknown values: b_k is the estimate obtained from the data; and $se(b_k)$ is the standard error of b_k that is computed from the dataset. Hence, I can compute the test statistic t_k by taking the ratio of b_k and $se(b_k)$.

The strategy to test the null hypothesis is as follows: I consider the degrees of freedom $(n - k)$, and the level of significance α . I compare the calculated t -value (obtained from equation (7.10) with the tabulated t -value (given at Appendix B). I compare the absolute value of t_k with the tabulated value. If the calculated value of t_k exceeds the tabulated value, I reject the null hypothesis. If the calculated value of t_k is less than the tabulated value, I *do not* reject the null hypothesis.

The critical value of t is defined as

$$P\{|t_k| > t_{n-k;\alpha/2}\} = \alpha$$

The critical value for $\alpha = 0.05$ for very large samples is 1.96. Thus, at the 5% level of significance the null will be rejected if

$$|t_k| > 1.96$$

Since under the alternative hypothesis, we have values for β_k on both sides of β_k^0 ($H_0: \beta_k = \beta_k^0$) the test is referred to as a two-tailed test. In case we have the null hypothesis as $H_0: \beta_k \leq \beta_k^0$ with an alternative hypothesis $H_1: \beta_k > \beta_k^0$, we reject the null hypothesis if t_k is too large. However, large negative values for t_k does not lead to rejection of the null hypothesis. This is known as one-sided test and the critical value is determined by

$$P\{t_k > t_{(n-k);\alpha}\} = \alpha$$

The critical value for $\alpha = 0.05$ for such a one-tailed test is 1.64. Thus, at the 5% level of significance the null will be rejected if

$$t_k > 1.64$$

Most of the standard statistical software reports the t -ratio as point estimates divided by its standard error. The t -ratio refers to the t statistics that is computed to test the hypothesis that $\beta_k = 0$. Rejection of the null hypothesis implies that the estimate b_k differs significantly from 0; or that the corresponding regressor x_{ik} has significant impact on y_i .

7.3.2 Confidence Interval

A confidence interval is the interval consisting of all values for β_k^0 for which the t -test does not reject the null hypothesis that $\beta_k = \beta_k^0$. It provides a range of values for true β_k given the estimate b_k and its standard error. Thus, for the given probability $(1 - \alpha)$ it implies that

$$-t_{(n-k);\alpha/2} < \frac{b_k - \beta_k^0}{se(b_k)} < t_{(n-k);\alpha/2} \quad \dots (7.11)$$

This can be written as

$$[b_k - t_{(n-k);\alpha/2} se(b_k)] < \beta_k < [b_k + t_{(n-k);\alpha/2} se(b_k)] \quad \dots (7.12)$$

Finally, using standard normal approximation, a 95% confidence interval for β_k is given by

$$[b_k - 1.96 se(b_k)] < \beta_k < [b_k + 1.96 se(b_k)] \quad \dots (7.13)$$

This implies that, in repeated sampling, in 95% cases, these intervals will contain true value of β_k .

7.3.3 F-Test

Almost all the statistical packages provide the results for the joint test of significance of the regression coefficients. It tests the hypothesis that all the coefficients except the intercept are equal to zero. The null here is that J of the k coefficients ($J < k$) are equal to zero.

$$H_0: \beta_{k-J+1} = \dots = \beta_k = 0 \quad \dots (7.14)$$

The alternative hypothesis is that at least one of these coefficients is not equal to zero. The test procedure applied here involves comparing the Residual Sum of Squares of the full model denoted by RSS_1 with Residual Sum of Squares (RSS) of the restricted model (model with the last J regressor omitted) denoted by RSS_0 . A test statistic for this can be obtained by using the following rationale.

Under the null hypothesis, and the assumptions discussed in Section 6.5 of the previous Unit, we have

$$\chi_J^2 \sim \frac{RSS_0 - RSS_1}{\sigma^2} \quad \dots (7.15)$$

From the null hypothesis, we know that $(RSS_0 - RSS_1)$ and s^2 (which is an estimator of σ^2) are independent and hence the test statistic can be defined as

$$F = \frac{\frac{(RSS_0 - RSS_1)}{J}}{\frac{s^2}{(n-k)}} \quad \dots (7.16)$$

The test statistic F is distributed as F with J and $(n-k)$ degrees of freedom. This is denoted as $F_{(n-k)}^J$. Hence, the test is also known as F -test. Using the definition of R^2 we can write (7.16) as

$$F = \frac{\frac{(R_1^2 - R_0^2)}{J}}{\frac{(1 - R_1^2)}{(n-k)}} \quad \dots (7.17)$$

where R_1^2 and R_0^2 are the goodness-of-fit measures for the unrestricted model and the restricted model respectively. The rejection of the null hypothesis is possible only in the case of large values of the test statistic. Although the alternative hypothesis is two-sided, based on the distribution, the critical values $F_{n-k; \alpha}^J$ of F is one-sided and is defined as

$$P\{F > F_{(n-k); \alpha}^J\} = \alpha, \text{ for the level of significance } \alpha.$$

A special case of F -test is where we test the null hypothesis that all the partial slope coefficients are zero, i.e., $H_0: \beta_2 = \beta_3 = \dots = \beta_k = 0$. In this case, the test statistic is given as

$$F = \frac{\frac{(RSS_0 - RSS_1)}{(k-1)}}{\frac{RSS_1}{(n-k)}} \quad \dots (7.18)$$

where RSS_0 is the RSS of the restricted model and RSS_1 is the RSS of the unrestricted model. Further, in this case, since in the restricted model only the intercept term is present, the test statistic using R^2 values is given as

$$F = \frac{\frac{R^2}{(k-1)}}{\frac{(1-R^2)}{(n-k)}} \quad \dots (7.19)$$

Here, F is used to denote the test statistic expressed as a simple function of R^2 . This implies that if the test based on F fails to reject the null hypothesis, we can infer that the model performs poorly. However, if the null hypothesis is rejected, we cannot infer that the model performs better.

Check Your Progress 2

- 1) Which ‘statistic’ is used to test for the significance of estimated individual regression coefficients? How is the test result arrived at?

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- 2) How is the rejection of a null hypothesis interpreted?

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- 3) Define the ‘confidence interval’ of a parameter. What does it imply?

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- 4) When is the F -statistic used? Indicate its expression in terms of R^2 stating how the interpretation is made depending on its value obtained for a sample.

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7.4 TESTING LINEAR RESTRICTIONS

The t -test discussed in Sub-Section 7.3.1 involved restriction on one parameter only. In practise, however, we come across situations which requires linear restrictions on more than one parameter such as $\beta_2 + \beta_3 + \dots + \beta_k = 1$.

For example, in the case of the Cobb-Douglas production function (which can be written as $\ln Y = \beta_0 + \beta_K \ln K + \beta_L \ln L + u$) we can impose the restriction $\beta_K + \beta_L = 1$ for having constant returns to scale. Such linear restrictions for some scalar q and a K -dimensional vector r can be written as

$$H_0: r_1\beta_1 + \dots r_K\beta_K = r'\beta = q \quad \dots (7.20)$$

Here we can test the hypothesis that $r'\beta$ is BLUE for $r'\beta$. The variance is $V\{r'\beta\} = r'V\{\beta\}r$. Now if we replace σ^2 with its estimate s^2 in $V\{\beta\}$ we get estimated covariance matrix $\hat{V}\{\beta\}$. This gives us the standard error of the linear combination $r'\beta$ as $se(r'\beta) = s\sqrt{r'\hat{V}\{\beta\}r}$. As both β and $r'\beta$ follow normal distribution, we have

$$\frac{r'\beta - r'\beta}{se(r'\beta)} \sim t_{n-k} \quad \dots (7.21)$$

This is similar to (7.3.1). Hence, the test statistic can be written as

$$t_k = \frac{r'\beta - q}{se(r'\beta)} \quad \dots (7.22)$$

If the absolute value of t is greater than 1.96, it leads us to reject the null hypothesis at 5% level of significance.

7.4.1 Wald Test

In general, we come across the hypothesis which is a combination of the two null hypotheses and a set of J linear restriction on the coefficients of the model. The restriction in such cases is given as

$$R\beta = q \quad \dots (7.23)$$

where R is $J \times K$ matrix with an assumption that it is of full rank equal to the number of rows and q is a J -dimensional vector. For example, let us suppose a model with a set of restrictions

- $\beta_2 + \beta_3 + \dots + \beta_k = 1$
- $\beta_2 = \beta_3$.
- $J = 2$
- $R = \begin{bmatrix} 0 & 1 & 1 & \dots & \dots & 1 \\ 0 & 1 & -1 & 0 & \dots & 0 \end{bmatrix}$
- $q = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Estimating the model with the above restrictions is possible by using (7.23). However, in some cases, estimating the model with these restrictions may not be possible under the null hypothesis. In such cases the result that

$$Rb \sim N(R\beta, \sigma^2, R(X'X)^{-1}R') \quad \dots (7.24)$$

can be used to formulate a quadratic form of equation.

It will be like (7.25) below which follows a Chi-squared distribution under the null hypothesis.

$$\xi = \frac{(Rb-q)'(R(X'X)^{-1}R')^{-1}(Rb-q)}{\sigma^2} \sim \chi_f^2 \quad \dots (7.25)$$

On replacing the unknown σ^2 in (7.25) by its estimate s^2 , we get a new statistic following a Chi-squared distribution. This statistic is referred to as the Wald Test. This test is based on the notion that the unrestricted model should 'approximately' satisfy the restrictions if they are true. Note that the Wald test provides the way for testing the restrictions without even estimating the restricted model.

7.4.2 Lagrange-Multiplier Test

Lagrange-Multiplier Test or the *LM*-test is used for testing the hypotheses expressed as one or more constraints on parameter values estimated under the likelihood framework. Unlike the Wald test, which requires estimation of the unrestricted model, *LM* -test requires estimation of the restricted parameter only. It is formulated to test whether the Lagrange multipliers which enforce the restrictions are significantly different from zero.

Let us suppose that $L(\theta)$ is a log-likelihood function of $(K \times 1)$ parameter vector θ and the score function and information matrix are given by

$$q(\theta) = \frac{\partial L(\theta)}{\partial \theta} \quad \dots (7.26)$$

$$I(\theta) = -E \left[\frac{\partial^2 L(\theta)}{\partial \theta \partial \theta'} \right] \quad \dots (7.27)$$

Let $\tilde{\theta}$ be the maximum likelihood estimator of θ subject to $(r \times 1)$ constraints $h(\theta) = \mathbf{0}$. Now the Lagrangian function is given by

$$\mathcal{L} = L(\theta) - \lambda' h(\theta) \quad \dots (7.28)$$

where λ is the $(r \times 1)$ vector of Lagrange multipliers. Now, the first-order conditions for $\tilde{\theta}$ are

$$\frac{\partial \mathcal{L}}{\partial \theta} = q(\tilde{\theta}) - H(\tilde{\theta})\tilde{\lambda} = \mathbf{0}$$

and

$$\frac{\partial \mathcal{L}}{\partial \lambda} = h(\tilde{\theta}) = \mathbf{0} \text{ where } H(\theta) = \partial h(\theta)' / \partial \theta.$$

Solving, we get the Lagrange Multiplier test statistic as

$$LM = \tilde{q}'\tilde{I}^{-1}\tilde{q} = \tilde{\lambda}'\tilde{H}'\tilde{I}^{-1}\tilde{H}\tilde{\lambda} \quad \dots (7.29)$$

where $\tilde{q} = q(\tilde{\theta})$, $\tilde{I} = I(\tilde{\theta})$ and $\tilde{H} = H(\tilde{\theta})$.

The term $\tilde{q}'\tilde{I}^{-1}\tilde{q}$ represents the 'score form' of the test statistic, whereas $\tilde{\lambda}'\tilde{H}'\tilde{I}^{-1}\tilde{H}\tilde{\lambda}$ represents the 'Lagrange Multiplier form' of the test statistic. In case we have an unrestricted MLE of θ , the score function $q(\theta) = \mathbf{0}$. However, it is not equal to zero in the case of restricted MLE($\tilde{\theta}$). If this constraint is true, we

would expect the values of $\tilde{\theta}$ and $\tilde{\lambda}$ to be small. Thus, large value of these two would lead to rejection of the null hypothesis $H_0: h(\theta) = \mathbf{0}$. If the constraint imposed under the null hypothesis is satisfied, the large sample LM statistic converges to chi-square distribution with $(k - r)$ degrees of freedom.

7.4.3 Likelihood Ratio Test

Likelihood ratio test or the LR -test enables us to test whether adding a set of new variables to the existing model increases the explaining capacity of the model. We saw that Wald test requires the estimation of only the unrestricted model, whereas, the LM -test requires the estimation of the restricted model. Unlike these two tests, the likelihood ratio test or LR -test requires estimation of both restricted as well as the unrestricted models.

Let us suppose that θ is a vector of parameter to be estimated having some restriction specified as the null hypothesis H_0 . The unrestricted and restricted maximum likelihood parameters of θ are denoted by $\hat{\theta}_u$ and $\hat{\theta}_R$ respectively. Under H_0 , the large sample distribution of $-2\ln\lambda$ is chi-square with degrees of freedom equivalent to the number of restrictions imposed, i.e., J . The LR test statistic is given by

$$LR = 2(\ln L(\hat{\theta}_u) - \ln L(\hat{\theta}_R)) \sim \chi_J^2 \quad \dots (7.30)$$

As said above, the LR -test enables us to test whether adding a set of new variables to the existing model increases the explaining capacity of the model. Using LR -test to test the simple null hypothesis against alternative hypothesis is not possible. This is because the degrees of freedom equals the number of parameters reduced as a consequence of imposing restrictions and in the case of testing the simple null hypothesis against alternative hypothesis, this would be equal to zero. Further, LR -test cannot be applied to test one distributional assumption against another as the likelihood functions of the two with different distributions will be unrelated.

7.4.4 Empirical Exercises

1) Suppose we are interested in finding out if the performance of a company, a crucial factor in determining the salary of the CEO. We have the model where the salary of the CEO is a function of the annual sales (*sales*), return on equity (*roe*), and return on stock (*ros*) as

$$\text{Salary} = \beta_1 + \beta_2 \text{sales} + \beta_3 \text{roe} + \beta_4 \text{ros} + \varepsilon_i$$

The following result is obtained:

$$\widehat{\text{salary}} = 4.3117 + 0.2803(\text{sales}) + 0.0174(\text{roe}) + 0.0002(\text{ros}) + u$$

$$(0.3154) \quad (0.0353) \quad (0.0041) \quad (0.0005)$$

$$R^2 = 0.2826, \bar{R}^2 = 0.2721 \quad n = 209$$

(a) Provide the test for the significance of the explanatory variables (i.e., $H_0: \beta_2 = 0; \beta_3 = 0$ and $\beta_4 = 0$).

(a) To test the hypotheses $H_0: \beta_2 = 0; \beta_3 = 0$ and $\beta_4 = 0$, t statistic is calculated for each of the coefficients as follows.

(i) $H_0: \beta_2 = 0$

$$t_{sales} = \frac{b_k - \beta_k^0}{se(b_k)} = \frac{0.2803 - 0}{0.0353} = 7.9405$$

(ii) $H_0: \beta_3 = 0$

$$t_{roe} = \frac{b_k - \beta_k^0}{se(b_k)} = \frac{0.0174 - 0}{0.00409} = 4.2543$$

(iii) $H_0: \beta_4 = 0$

$$t_{ros} = \frac{b_k - \beta_k^0}{se(b_k)} = \frac{0.0002 - 0}{0.0005} = 0.4401$$

As the value for both t_{sales} and t_{roe} is greater than the critical value of 1.96, we reject the null hypothesis $H_0: \beta_2 = 0$ and $H_0: \beta_3 = 0$. Hence, we can say that both *sales* and *roe* are having positive influence on the salary of the CEO. The value for t_{ros} being small we cannot reject the null that $H_0: \beta_4 = 0$. Hence, it is statistically not significant.

(b) Test the overall significance of the model (i.e., $H_0: \beta_2 = \beta_3 = \beta_4 = 0$).

The overall significance of the model ($H_0: \beta_2 = \beta_3 = \beta_4 = 0$) can be tested by computing the F statistic

$$F = \frac{R^2/k - 1}{(1 - R^2)/(n - k)} = \frac{0.2826/(4 - 1)}{(1 - 0.2826)/(209 - 4)} = 26.91$$

Based on the large F statistic obtained, we get p -value equal to 0. This means we can reject the null hypothesis ($H_0: \beta_2 = \beta_3 = \beta_4 = 0$) that the performance of the company has no effect on the salary of the CEO at all significance level or that the overall model is significant.

(2) Suppose we have the following model

$$wage = \beta_1 + \beta_2 education + \beta_3 experience + \beta_4 tenure + \beta_5 age + \varepsilon_i$$

Now, we decide to drop *tenure* from the model as in many cases it would be equal to *experience*. Also, we decide in favour of dropping *age* as it may be highly correlated with *experience*. How would you test that dropping these two variables is acceptable?

Here, the null and the alternative is of the following nature

$$H_0: \beta_3 = \beta_4 = 0$$

$H_1: H_0$ is not true

The restricted model corresponding to H_0 will be

$$\text{wage} = \beta_1 + \beta_1 \text{education} + \beta_4 \text{experience} + u$$

$$\text{wage} = \beta_1 + \beta_2 \text{education} + \beta_4 \text{experience} + \varepsilon_i$$

To test the null, both the unrestricted and the restricted models are estimated. Based on the residual sum of the squares (RSS) obtained after estimation of the two models S_0 and S_1 respectively, the F statistic computed using

$$F = \frac{(RSS_0 - RSS_1)/J}{RSS_1/(n-k)}$$

Depending on the F statistic, p -value is determined. If we are able to reject the null hypothesis, we can conclude that *tenure* and *age* are jointly significant and hence dropping them is not acceptable. On the other hand, if we fail to reject the null, it would mean that they are not jointly significant and excluding the variable is acceptable.

Check Your Progress 3

- 1) Explain why we need to impose ‘linear restrictions’ in a regression model. Give an example.

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- 2) What is Wald’s test? On what rationale is it based? What is its advantage?

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- 3) In what way is the LM test different from the Wald’s test?

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- 4) What specific purpose does the LR Test serve? How is this test different from that of Wald’s and the LM tests?

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- 5) Why is it not possible to apply the LR test in the case of testing a simple null hypothesis against an alternative hypothesis?

7.5 LET US SUM UP

Evaluation of a regression model could be made through the coefficient of determination, which indicates the percentage variation in dependent variables that be explained by the independent variables. Hypothesis testing in multiple regression models is carried out to examine if a finding is sufficiently close to the stated hypothesis associated with the unknown population parameter vector β . After providing an outline of the coefficient of determination R^2 , details of the two types of tests. Commonly used tests, viz., the t -test and the F -Test are discussed in this unit. In practise, we come across hypotheses which require linear restrictions on more than one coefficient. Testing the hypotheses in such cases can be done either by reparameterization or by using alternative tests like the Wald, Lagrange Multiplier or the Likelihood ratio tests. These tests have been detailed out in the last section of the unit.

7.6 KEY WORDS

Lagrange Multiplier (LM) Test	: LM test is used for testing the hypotheses expressed as one or more constraints on the parameter values estimated under the likelihood framework.
Biased Estimator	: An estimator whose expectation, or sample mean, is different from the population value it is supposed to be estimating.
Likelihood Ratio Statistic	: A statistic that can be used to test single or multiple hypotheses when the constrained and unconstrained models are estimated by the maximum likelihood method.
Maximum Likelihood Estimation (MLE)	: A broadly applicable estimation method where the estimates are chosen to maximize the log-likelihood function.

- Quadratic Functions** : It refers to the functions that contain squares of one or more explanatory variables; they capture diminishing or increasing effects on the dependent variable.
- Restricted Model** : It is a model in hypothesis testing obtained after imposing all the restrictions required under the null.

7.7 SOME USEFUL BOOKS

- Greene, W. H., 2003, “*Econometric Analysis*.” Prentice Hall, New Jersey, USA. 5thed., pp. 484-496.
- Maddala, G. S., 1992, “*Introduction to Econometrics*”, Macmillan Publishing Company, New York. 2^d ed., pp. 127–177.
- Verbeek, M. 2004, “*A Guide to Modern Econometrics*”. John Wiley & Sons Ltd, England. 2nd ed., pp. 8–49.
- Wooldridge, Jeffrey M. 2000, “*Introductory Econometrics*”, South-Western College Publishing, 2000. Chapter 3; pp. 66–69.

7.8 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) It is defined as the ratio of Explained Sum of Squares (ESS) to Total Sum of Squares (TSS). Its expression is given by equation (7.3): $R^2 = \frac{ESS}{TSS} = \frac{\sum(\hat{Y}_i - \bar{Y})^2}{\sum(Y_i - \bar{Y})^2}$.
- 2) R^2 measures the proportion of the variation in Y which is explained by the multiple regression equation. Therefore, R^2 is often used as a goodness-of-fit statistic to compare the validity of regression results under alternative specifications of the independent variables in the model. Another reason is that the coefficient of determination (R^2) reflects the explanatory power of a regression model. A higher value of R^2 implies higher explanatory power of the underlying regression model compared to some other model. Thus, R^2 can be used as a measure of goodness of fit in multiple regressions.
- 3) $R^2 = 1 - \frac{\sum \hat{u}_i^2}{\sum(Y_i - \bar{Y})^2}$. This expression conveys that ‘as RSS decreases, R^2 increases’.
- 4) It is a fact that addition of a new explanatory variable invariably increases the value of R^2 . However, we cannot trust the inferences drawn from such models! This is because, there is a decrease in the degrees of freedom of RSS when we add explanatory variables to a model. A way out of this situation is to use adjusted- R^2 given by $\bar{R}^2 = 1 - (1 - R^2) \frac{(n-1)}{(n-k)}$ where n is number of observations in the regression model and k is number of

parameters in the equation. This approach eliminates the dependence of goodness of fit on the number of independent variables in the model.

- 5) One, when new variables are added to a regression model, while R^2 always increases, $\bar{R}^2$ may rise or fall. Second, The use of $\bar{R}^2$ eliminates the incentives to include numerous variables in a model. Third, $\bar{R}^2$ is always less than or equal to R^2 . Fourth, $\bar{R}^2$ can be negative also while R^2 is always positive. For these reasons, $\bar{R}^2$ is regarded as a more desirable goodness of fit measure compared to R^2 .

Check Your Progress 2

- 1) For testing $H_0: \beta_k = \beta_k^0$, we use the statistic $t_k = \frac{b_k - \beta_k^0}{se(\beta_k)}$. We reject the null hypothesis if the absolute value of the test statistic $|t_k|$ is very large. For large samples, at 5 % level of significance the null will be rejected if $|t_k| > 1.96$
- 2) The t -ratio refers to the t statistic that is computed to test the hypothesis that $\beta_k = 0$. Rejection of the null means that the estimate b_k differs significantly from 0 or that the corresponding regressor x_{ik} has significant impact on y_i .
- 3) A confidence interval is the interval consisting of all values for β_k^0 for which the t -test does not reject the null hypothesis that $\beta_k = \beta_k^0$. It provides a range of values for true β_k given the estimate b_k and its standard error. Thus, for the given probability $(1 - \alpha)$ it implies that $-t_{n-k; \alpha/2} < \frac{b_k - \beta_k^0}{se(b_k)} < t_{n-k; \alpha/2}$. If the level of significance is chosen as 5%, then the confidence interval implies that, in repeated sampling, in 95% cases, these intervals will contain true value of β_k .
- 4) The F -test is used to test for the joint significance of the estimated regression model. Specifically, it tests a hypothesis like: $H_0: \beta_2 = \beta_3 = \dots = \beta_K = 0$. Using the R^2 values of two regressions, viz., the full model denoted by RSS_I with that of the squared residuals of the restricted model (model with the last J regressor omitted) denoted by RSS_0 , the F -statistic is defined as: $F = \frac{(RSS_0 - RSS_I)/J}{RSS_I/(n-k)}$. In terms of R^2 , it is defined as: $\frac{R^2/(k-1)}{(1-R^2)/(n-k)}$.

Check Your Progress 3

- 1) The t -test considers restriction on one coefficient only like $\beta_k = 0$. In practise, we come across economic hypothesis which requires linear restrictions on more than one coefficients such as $\beta_2 + \beta_3 + \dots + \beta_k = 1$. For instance, in the case of the Cobb-Douglas production function (which is written in logarithmic form in a linear regression model), the constant returns to scale is represented by the sum of all the coefficients associated with the inputs as equal to one.
- 2) A statistic as in Equation (7.26), used to test J linear restrictions such as $H_0: R\beta = q$, is referred to as the Wald's test. It is based on the notion that

the unrestricted model should ‘approximately’ satisfy the restrictions if they are true. Its advantage is that it provides a method for testing the restrictions without requiring the estimation of the restricted model.

- 3) Wald’s test requires the estimation of the ‘unrestricted model’. It does not require the estimation of the ‘restricted model’. The LM test, on the other hand, requires the estimation of the ‘restricted parameter’ only.
- 4) Likelihood ratio test or the *LR*-test enables us to test whether adding a set of new variables to the existing model increases the explaining capacity of the model. Unlike the Wald’s and the LM test, which requires the estimation of either the restricted or the unrestricted models, the likelihood ratio test requires the estimation of both the restricted as well as the unrestricted models.
- 5) This is because in the LR test, the degree of freedom equals the number of parameters reduced as a consequence of imposing the restrictions. In the case of testing the simple null hypothesis against alternative hypothesis, this would be equal to zero.



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